

1 All independent coefficient faces attached to the original theta source

This construction attaches the independent coefficient masks of the original mixed-support algebra to the actual theta complexes. It retains the original outer source label and leg mask independently of the coefficient mask. A further slot-local leg diagram and its exact comparison with that outer diagram are constructed explicitly. The source-level maps, their cohomology, and the full arithmetic boundary of the finite section are computed below. Every tensor operation already present in the ambient programme keeps its original tensor variables; the finite direct sums in this section are explicitly coefficient-slot observations.

1.1 The original source and its full finite arithmetic sections

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+) : \sup_{x>0} x^b |D^j F(x)| < \infty \quad (b \in \mathbb{Z}, j \geq 0)\}, \\ D &= -x\partial_x, \quad \widehat{\phi}(\nu) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \nu} dx, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q. \end{aligned} \tag{MCF1}$$

The letter q in (MCF1) denotes the quotient map; a coefficient Frobenius prime power will be denoted q_0 below. Put

$$\begin{aligned} U &= \{s : 0 < \Re s < 1, \Re s \neq \tfrac{1}{2}\}, \quad T_U Q = \bigcup_p \ker(p(D) : Q \rightarrow Q), \\ \mathcal{B}_U &= q^{-1}(T_U Q), \quad C_U = [V \xrightarrow{\Theta} \mathcal{B}_U], \end{aligned} \tag{MCF2}$$

where p ranges over all monic polynomials with roots in U , including 1. The original source calculation proves injectivity of Θ , $D\Theta = \Theta D$, and injectivity of every nonzero polynomial $p(D)$ on V and $\mathcal{B}$. We use its exact range identities

$$\begin{aligned} p(D)V &= \ker(j_p H), \quad p(D)\mathcal{B} = \ker(j_p \mathcal{M}), \\ \mathcal{M}F(s) &= \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \mathcal{M}\Theta\phi = gH_\phi, \quad g = 2\xi, \\ H_{P(D)\phi_*} &= P, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}. \end{aligned} \tag{MCF3}$$

These are the original arithmetic identities (AG1) and (CAU1), with their full-plane Euler inverses. The attachment below uses their stated domains; it does not introduce a different test-function completion. For example, the last identity follows from

$$M_+ \phi_*(s) = \tfrac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2), \quad H_\phi = \frac{2\pi^{s/2} M_+ \phi}{s(s-1) \Gamma(s/2)}.$$

The displayed Mellin formula is obtained by integrating the two Gaussian monomials and using $\Gamma(z+1) = z\Gamma(z)$; the apparent exceptional values are handled by the original entire continuation. The moments of ϕ_* are zero because $\phi_*(0) = 0$ and $4\pi^2 \int x^4 e^{-\pi x^2} dx - 6\pi \int x^2 e^{-\pi x^2} dx = 3 - 3 = 0$.

For a nonempty finite full-order packet in U , write

$$\begin{aligned} h(s) &= \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad m_\rho = \text{ord}_\rho g, \quad d_h = \deg h, \\ E_h &= \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h, \quad \varepsilon_h = v_h^{-1}, \\ F_h &= S_h^{\mathcal{B}} \Theta \phi_*, \quad P_h(u) = r_h(\varepsilon_h u), \quad s_h u = P_h(u)(D)F_h, \quad \sigma_h = q s_h. \end{aligned} \tag{MCF4}$$

Here r_h is the unique polynomial representative of degree less than d_h , and $S_h^{\mathcal{B}}$ is the inverse of $h(D)$ on the exact range in (MCF3). Every actual multiplicity is retained, so v_h is nonzero at each packet centre and v_h is a unit in the full local algebra. Every packet index and directed limit in this module also includes the empty packet $h = 1$, with $F_1 = \Theta\phi_*$, $E_1 = 0$, $s_1 = \sigma_1 = \ell_1 = 0$, and $\mathcal{B}_1 = \Theta V$. Its maps into other packet stages are zero on E_1 and the original inclusion on ΘV . No inverse unit at this zero-vector-space stage is used. The definition of F_1 is forced by $1(D)F_1 = \Theta\phi_*$ and gives $\mathcal{M}F_1 = g$. Thus the directed systems are defined even when there is no nonempty packet in U . The equalities

$$\begin{aligned} h(D)F_h &= \Theta\phi_*, & \mathcal{M}F_h &= v_h, & J_h s_h &= 1, & J_h &= j_h \mathcal{M}, & J_h \Theta &= 0, \\ h(D)s_h u &= \Theta(P_h(u)(D)\phi_*), \\ Ds_h - s_h A_h &= \Theta\phi_* \ell_h, & \ell_h(u) &= [s^{d_h-1}]P_h(u) \end{aligned} \quad (\text{MCF5})$$

have the original source types. To verify the last equality in full, $sP_h(u) - P_h(A_h u)$ has zero h -jet, degree at most d_h , and leading coefficient $\ell_h(u)$. Monicity gives $sP_h(u) - P_h(A_h u) = h\ell_h(u)$. Apply this polynomial in D to F_h and use the first equality. The other identities follow directly from (MCF3)–(MCF4) and retain the entire multiplier v_h .

For completeness the finite arithmetic image is exactly $\sigma_h E_h = Q[h(D)]$. If $h(D)F = \Theta\phi$, the assignment $[F] \mapsto j_h H_\phi$ is well-defined because replacing F by $F + \Theta\psi$ replaces ϕ by $\phi + h(D)\psi$. A zero jet makes $\phi = h(D)\psi$ by (MCF3), and injectivity of $h(D)$ gives $F = \Theta\psi$. Every jet represented by a polynomial P is obtained from $S_h^{\mathcal{B}}\Theta(P(D)\phi_*)$, since $j_h(gP) = 0$. Thus this assignment is an isomorphism. Applied to $s_h u$ it gives $\varepsilon_h u$ by (MCF5), an invertible change of full jet coordinates. In particular σ_h is injective and has the asserted image.

1.2 The independent coefficient face category and its exact quotient

Fix $n \geq 1$ and $I_n = \{0, \dots, n-1\}$. For every subset $A \subseteq I_n$, including the empty subset, put

$$C_U[A] = \bigoplus_{i \in A} C_U = [V^A \xrightarrow{(\phi_i) \mapsto (\Theta\phi_i)} \mathcal{B}_U^A], \quad E_h[A] = E_h^A. \quad (\text{MCF6})$$

Empty direct sums are zero vector spaces. If $A \subseteq B$, define z_{AB} by the identity in each old coordinate and by the typed zero in every coordinate of $B \setminus A$, in both degrees and in $E_h[A]$. Then $dz_{AB} = z_{AB}d$ coordinate by coordinate, and $z_{BC}z_{AB} = z_{AC}$. This is a functor from the inclusion category $\mathcal{P}(I_n)$ to complexes. The complete face diagram is retained; no colimit is taken that would identify its different masks.

For a complex vector space X , set $G(X) = \{\tau\} \sqcup \{x^\bullet : x \in X\}$, with the original supported addition and scalar action. The degree- j mixed carrier is

$$\mathcal{D}_n^j = \prod_{i \in I_n} G(C_U^j) \simeq \prod_{A \subseteq I_n} \{A\} \times (C_U^j)^A. \quad (\text{MCF7})$$

The empty term on the right is one point, corresponding to the all-absent tuple. The bijection sends each present coordinate to its original amplitude and records every present zero in A . Its inverse puts those amplitudes in the coordinates of A as supported values and puts τ in the other coordinates. Thus all 2^n masks, including the proper nonempty faces, are present. A face arrow z_{AB} changes the recorded mask from A to B and fills the new coordinates with supported zeros. It is an arrow between labelled faces, not an identification of them.

The differential preserves the original mask of its input and applies Θ independently in every present coordinate. A source vector annihilated by its amplitude differential reaches the supported zero at the same mask. In the internal split cycle and boundary construction, $\epsilon(A, x) = (A, 0)$ and

$Z = \text{Eq}(d, \epsilon d)$; boundary relations are the coequalizer of b and ϵb within each labelled fibre. Consequently the exact cohomology and quotient are

$$\begin{aligned} H^0(C_U[A]) &= 0, & H^1(C_U[A]) &= (T_U Q)^A, \\ \mathcal{D}_n^1 &\longrightarrow \prod_i G(T_U Q), & F_i^\bullet &\longmapsto (qF_i)^\bullet, \quad \tau \longmapsto \tau. \end{aligned} \quad (\text{MCF8})$$

Indeed Θ is injective in every active coordinate, and the image in degree one is exactly $(\Theta V)^A$. Two amplitudes at mask A are identified precisely when their coordinate differences belong to $(\Theta V)^A$. The fibre over the supported zero at A is $\{A\} \times (\Theta V)^A$; it is not combined with any other mask. More generally, choosing any lift F of an amplitude in $(T_U Q)^A$ gives its entire fibre as $F + (\Theta V)^A$ at that same mask.

The full finite arithmetic attachment is

$$\begin{array}{ccccc} E_h[A] & \xrightarrow{s_h^A} & \mathcal{B}_U^A & \xrightarrow{q^A} & (T_U Q)^A \\ z_{AB} \downarrow & & z_{AB} \downarrow & & \downarrow z_{AB} \\ E_h[B] & \xrightarrow{s_h^B} & \mathcal{B}_U^B & \xrightarrow{q^B} & (T_U Q)^B. \end{array} \quad (\text{MCF9})$$

All squares commute because every displayed map is applied independently to the unchanged old coordinates and is linear on the inserted zeros. Every $\sigma_h^A = q^A s_h^A$ is injective by (MCF5). In every active coordinate its exact Euler boundary is

$$(D^A s_h^A - s_h^A A_h^A)(u_i)_{i \in A} = (\Theta \phi_* \ell_h(u_i))_{i \in A}. \quad (\text{MCF10})$$

This is an equality of actual functions, whose primitive is $(\phi_* \ell_h(u_i))_{i \in A} \in V^A$. After the original quotient it proves $D^A \sigma_h^A = \sigma_h^A A_h^A$. Thus every proper coefficient face carries the same full arithmetic boundary calculation, with its mask still specified.

1.3 Independent slot-local leg presentations

There is also a slot-local copy of the original two-leg presentation at every active coefficient. This auxiliary indexing does not replace the original outer leg label: its exact attachment, including empty coefficient support, is (MCF34)–(MCF40) below. Its objects are

$$\lambda = (A, (S_i)_{i \in A}), \quad A \subseteq I_n, \quad \emptyset \neq S_i \subseteq \{+, -\}. \quad (\text{MCF11})$$

For empty A the family is empty. Order these objects by $\lambda \leq \lambda'$ when $A \subseteq A'$ and $S_i \subseteq S'_i$ for every $i \in A$. Thus each coefficient slot has four states: absent, plus only, minus only, or both. There are exactly 4^n slot-local masks.

At λ put

$$\begin{aligned} C_\lambda^0 &= \bigoplus_{i \in A} \left(\bigoplus_{+ \in S_i} V_+ \oplus \bigoplus_{- \in S_i} V_- \right), \\ C_\lambda^1 &= \mathcal{B}_U^A, \quad d_\lambda(\phi_i, \psi_i)_i = (\Theta \phi_i - \Theta \hat{\psi}_i)_i. \end{aligned} \quad (\text{MCF12})$$

A coordinate of a missing leg in the displayed formula is its typed zero; that convention evaluates the formula without adjoining that leg to the mask. The plus and target actions of $t \in \mathbb{C}[t]$ are D ; the minus action is $1 - D$. The Fourier identity $\widehat{D}\phi = (1 - D)\hat{\phi}$ proves that this is a complex of $\mathbb{C}[t]$ -modules. For $\lambda \leq \lambda'$, retain every old source leg and coefficient, insert typed zeros in the new legs and slots, and use zero insertion in degree one. Both differentials then agree term by term, proving a strictly composing cochain map. The actual supported arrow changes the label from λ to λ' and retains every newly supported zero as such.

The natural comparison with (MCF6) is the cochain map

$$p_\lambda : C_\lambda \longrightarrow C_U[A], \quad p_\lambda^0(\phi_i, \psi_i)_i = (\phi_i - \widehat{\psi}_i)_i, \quad p_\lambda^1 = 1. \quad (\text{MCF13})$$

It commutes with every zero-insertion transition. In a plus-only slot its degree-zero map is the identity, in a minus-only slot it is $-\widehat{}$, and in a joint slot it has the exact inverse coordinates

$$u = \phi - \widehat{\psi}, \quad v = \frac{1}{2}(\phi + \widehat{\psi}), \quad \phi = v + \frac{1}{2}u, \quad \psi = \widehat{v} - \frac{1}{2}\widehat{u}. \quad (\text{MCF14})$$

Fourier squares to the identity on V , so every displayed map and inverse has its stated domain. Write $J(\lambda) = \{i \in A : S_i = \{+, -\}\}$. The exact kernel and cohomology are

$$\begin{aligned} 0 \longrightarrow V^{J(\lambda)}[0] &\xrightarrow{\delta_\lambda} C_\lambda \xrightarrow{p_\lambda} C_U[A] \longrightarrow 0, \\ \delta(v)_i &= (v_i, \widehat{v}_i) \quad (i \in J(\lambda)), \\ H^0(C_\lambda) &= V^{J(\lambda)}, \quad H^1(C_\lambda) = (T_U Q)^A. \end{aligned} \quad (\text{MCF15})$$

Surjectivity of p^0 follows from the single-leg formulas and (MCF14). Its kernel in a joint slot is exactly the displayed diagonal, and its degree-one kernel is zero. Its differential on (u, v) is Θu . Injectivity of Θ now proves the cohomology statement directly. Thus the two-leg diagonal remains present at every relevant independent coefficient; it is not removed when the arithmetic degree-one quotient is observed.

The slot-local cohomology carrier is the following. The separate outer label is retained in (MCF34).

$$\widetilde{H}_n^1 = \{\tau_n\} \sqcup \coprod_{\substack{\lambda=(A,S) \\ A \neq \emptyset}} \{\lambda\} \times (T_U Q)^A. \quad (\text{MCF16})$$

Forgetting only the leg masks gives a pointed map to $\prod_i G(T_U Q)$. Over a specified tuple of coefficient mask A it has exactly $3^{|A|}$ labelled preimages, one for every nonempty S_i ; the amplitudes are fixed. Over the all-absent tuple its fibre is the single τ_n . A specified section takes the joint leg mask at every active coefficient. These formulas prove the exact relationship between the combined support diagram and the coefficient-mask observation; neither is silently used in place of the other. The maps s_h^A and σ_h^A of (MCF9) give their source and arithmetic amplitudes at every λ , with the same recorded coefficient and leg masks.

One can also lift each original one-leg complex into its selected leg presentation. At a single slot the degree-zero section of p_λ is

$$j_+(u) = u, \quad j_-(u) = -\widehat{u}, \quad j_{+-}(u) = (u/2, -\widehat{u}/2), \quad (\text{MCF17})$$

and degree one is the identity. They are $\mathbb{C}[t]$ -linear cochain maps by the Fourier identity. For the two inclusions into the joint slot their precise difference is

$$\text{inc}_+ j_+ - j_{+-} = \delta(u/2), \quad \text{inc}_- j_- - j_{+-} = -\delta(u/2) \quad \text{in degree zero}, \quad (\text{MCF18})$$

and zero in degree one. These retain the original diagonal; a claim of strict naturality for these three chosen sections would omit it. There is also an explicit complex-linear homotopy. The coherent original sections give $\mathcal{B}_U = \Theta V \oplus s_U E_U$, with $E_U = \varinjlim_h E_h$; define $t_U(\Theta\phi + s_U u) = \phi$. Then $t_U \Theta = 1$. The degree-minus-one maps $H_\pm(F) = \pm\delta(t_U F)/2$ obey $dH_\pm = 0$ and $H_\pm \Theta = \pm\delta/2$, so $dH_\pm + H_\pm d$ equals (MCF18). This is the exact homotopy of the chosen sections. No equivariance of t_U or $H_\pm$ is asserted. Existence of the coherent s_U and its original unit is checked in the divisor calculation below.

1.4 Arithmetic dilation on every face and its full source boundary

For $a > 0$ write $\mathcal{R}_a F(x) = F(x/a)$, with the same formula on V . Changing variables shows that it preserves the original spaces, preserves the value-zero condition, and multiplies the source integral by a . Direct substitution gives

$$\Theta \mathcal{R}_a = \mathcal{R}_a \Theta, \quad \widehat{\mathcal{R}_a \phi} = a \mathcal{R}_{1/a} \widehat{\phi}, \quad \mathcal{M} \mathcal{R}_a F = a^s \mathcal{M} F. \quad (\text{MCF19})$$

It commutes with D and hence preserves $T_U Q$ and its inverse image $\mathcal{B}_U$. On every combined face its action is

$$(\phi_i, \psi_i)_i \mapsto (\mathcal{R}_a \phi_i, a \mathcal{R}_{1/a} \psi_i)_i, \quad F_i \mapsto \mathcal{R}_a F_i. \quad (\text{MCF20})$$

The coefficient and leg masks are unchanged. The Fourier formula in (MCF19) makes the two differentials commute exactly. These actions also commute with every face inclusion and with p_λ , since

$$\mathcal{R}_a \phi - a \widehat{\mathcal{R}_{1/a} \psi} = \mathcal{R}_a (\phi - \widehat{\psi}).$$

On the diagonal the action is $v \mapsto \mathcal{R}_a v$, because its minus coordinate becomes $a \mathcal{R}_{1/a} \widehat{v} = \widehat{\mathcal{R}_a v}$. Thus (MCF15) is an equivariant exact sequence.

The finite arithmetic action is

$$A_{a,h} = M_{j_h(a^s)} : E_h \longrightarrow E_h, \quad A_{a,h}|_\rho = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho = M_{s-\rho}. \quad (\text{MCF21})$$

The complete nilpotent Taylor polynomial is displayed. Mellin and (MCF19) give $J_h \mathcal{R}_a = A_{a,h} J_h$. Dilation preserves $Q[h(D)]$, and J_h is the inverse of σ_h there by (MCF5). Therefore $\mathcal{R}_a \sigma_h = \sigma_h A_{a,h}$, an exact equality on the original arithmetic quotient.

The source defect has a full original primitive, rather than an assumed equivariant lift. Define

$$\begin{aligned} \chi_{a,h}(u) &= \mathcal{R}_a(P_h(u)(D)\phi_*) - P_h(A_{a,h}u)(D)\phi_* \in V, \\ \Psi_{a,h}(u) &= S_h^V \chi_{a,h}(u), \end{aligned} \quad (\text{MCF22})$$

where S_h^V is the inverse of $h(D)$ on $\ker(j_h H)$ from (MCF3). This inverse is applied on its proved domain: the Mellin calculation gives

$$H_{\chi_{a,h}(u)} = a^s P_h(u) - P_h(A_{a,h}u), \quad j_h H_{\chi_{a,h}(u)} = j_h(a^s) \varepsilon_h u - \varepsilon_h A_{a,h}u = 0. \quad (\text{MCF23})$$

The complete $\varepsilon_h = j_h(g/h)^{-1}$ occurs in both terms. Applying $h(D)$ to the two sides and using (MCF5), (MCF19), and $h(D)\Psi = \chi$, gives

$$\mathcal{R}_a s_h u - s_h A_{a,h} u = \Theta \Psi_{a,h}(u). \quad (\text{MCF24})$$

Indeed the difference after application of $h(D)$ is zero, and its injectivity on $\mathcal{B}$ proves the displayed equality. In every coefficient face this equality is applied separately to each u_i . In a combined leg presentation an actual degree-zero primitive is $j_{S_i}(\Psi_{a,h}(u_i))$ from (MCF17). Its differential is precisely the corresponding coordinate of (MCF24). The alternative single-leg and joint-leg primitives differ by the diagonal terms of (MCF18), which remain recorded in (MCF15).

The primitives themselves have the exact composition law

$$\Psi_{ab,h}(u) = \mathcal{R}_a \Psi_{b,h}(u) + \Psi_{a,h}(A_{b,h}u). \quad (\text{MCF25})$$

Expand $\mathcal{R}_{ab} s_h - s_h A_{a,h} A_{b,h}$ as $\mathcal{R}_a(\mathcal{R}_b s_h - s_h A_{b,h}) + (\mathcal{R}_a s_h - s_h A_{a,h}) A_{b,h}$, apply (MCF24), and cancel the injective Θ . This proves the law with the original dilation order, source action, and full arithmetic multiplier.

1.5 Packet enlargement with every coefficient face retained

For full-order packets $h \mid H$, put $b = H/h$. Their supports differ by whole local factors, so b and h are coprime. Let $\iota_{hH} : E_h \rightarrow E_H$ insert the old full jets and zero jets at every added centre. CRT gives this injective $\mathbb{C}[s]$ -linear map and the restriction left inverse. At old centres $v_H = b^{-1}v_h$ and $\varepsilon_H = b\varepsilon_h$. At added centres b vanishes to the full required order. Hence

$$r_H(\varepsilon_H \iota_{hH} u) = b r_h(\varepsilon_h u), \quad b(D)F_H = F_h, \quad s_H \iota_{hH} = s_h. \quad (\text{MCF26})$$

When $h = 1, b = H$ and the middle identity is exactly $H(D)F_H = \Theta\phi_* = F_1$; the first and third identities have zero domain and are zero-map identities. The highest-coefficient, action, and primitive identities below also have zero domain there. For a nonempty h , the first identity holds in every full local factor and both polynomials have degree less than $\deg H$; uniqueness of the remainder proves it. For the second identity, apply $h(D)$ and use the injectivity in (MCF3). Together they prove the third. Monicity of b preserves the highest coefficient of the remainders in the first identity, giving

$$\ell_H \iota_{hH} = \ell_h, \quad A_{a,H} \iota_{hH} = \iota_{hH} A_{a,h}, \quad \Psi_{a,H} \iota_{hH} = \Psi_{a,h}. \quad (\text{MCF27})$$

The action identity follows in the unchanged old jets and the zero added jets. The primitive identity follows from (MCF24), the section identity, and injectivity of Θ . All statements hold coordinatewise in $E_h[A]$ and commute with every coefficient and combined-leg face map. Thus the finite face attachments glue by literal source equalities. Their directed union supplies the coherent s_U used in (MCF18). The decomposition $\mathcal{B}_U = \Theta V \oplus s_U E_U$ follows from surjectivity onto $T_U Q$ and injectivity of each σ_h ; every vector and every finite relation occurs in a common full packet. The original all-polynomial calculation (CAU5)–(CAU13) proves that these packets exhaust $T_U Q$, retaining every local zero order.

1.6 The original negacyclic masks and coefficient Frobenius

At its finite arithmetic-algebra vertex, the complete independent carrier is the original

$$D_{E_h,n} = G(E_h)^n, \quad B_{h,n} = E_h[t]/(t^n + 1), \quad \pi_n(c) = \sum_{i=0}^{n-1} p(c_i)t^i. \quad (\text{MCF28})$$

The variable t here is the additional coefficient variable of this finite observation. The spectral variable in E_h is still s . The negacyclic product, its original unit, and its mask law are

$$(c \star d)_k = \bigoplus_{i+j=k} c_i d_j \oplus \bigoplus_{i+j=k+n} (-1)^{\bullet} c_i d_j, \quad 1_D = (1^{\bullet}, \tau, \dots, \tau), \quad (\text{MCF29})$$

$$\text{mask}(c + d) = \text{mask}(c) \cup \text{mask}(d), \quad \text{mask}(c \star d) = \text{mask}(c) + \text{mask}(d) \pmod{n}.$$

For the product, an output is supported exactly when at least one pair of supported input indices sums to it. Wrap signs are supported and a nonempty sum of supported summands remains supported even when its coefficient cancels. This proves the stated law with every mixed face. Monic remainder in t gives the unique coefficients of $\pi_n(c)$; the mask then distinguishes each coefficient zero from absence. Thus (mask, π_n) is injective. Its facewise source and cohomology attachment is exactly (MCF9), or (MCF16) when leg masks are also retained. The product in (MCF29) belongs to this finite algebra vertex. The source maps s_h^A have their stated linear types; no unproved ring multiplication on the original source is inserted.

Let q_0 be an odd prime power with $\gcd(q_0, n) = 1$, and put $\kappa(i) = q_0 i \pmod{n}$ and $\epsilon_i = (-1)^{\lfloor q_0 i/n \rfloor}$. The coefficient Frobenius lift on the entire independent object is

$$c_i \mapsto \epsilon_i^{\bullet} c_i \quad \text{in slot } \kappa(i), \quad A \mapsto \kappa(A). \quad (\text{MCF30})$$

An absent coordinate stays absent. On a combined label it sends S_i to $S'_{\kappa(i)} = S_i$ and multiplies every present source-leg and target amplitude in that coefficient by ϵ_i . It commutes with the differential because it uses the same coefficient sign in each term of that differential. It commutes with face inclusion, the maps p_λ , s_h^A , σ_h^A , and all arithmetic dilations by coordinate permutation and complex linearity.

On the integral coefficient ring $\mathbb{Z}[t]/(t^n + 1)$ this is the matrix of $t \mapsto t^{q_0}$. It is well-defined since $(t^{q_0})^n = -1$, and choosing r with $q_0^r \equiv 1 \pmod{2n}$ proves its r th power is the identity. Reduction to $\mathbb{F}_{q_0}$ is exactly coefficient Frobenius; extension to $\mathbb{C}$ retains the same integral signed-permutation matrix. On supported monomials, reducing the exponent before or after substitution gives the same wrap sign, since q_0 is odd. Distributivity and the absent cases therefore prove multiplicativity on (MCF29), before synchronization. In particular its support action is the exact permutation in (MCF30), rather than an observation of one global Boolean support bit.

The synchronization arrow is also retained explicitly:

$$\mathbf{E}_n = (1^\bullet, e, \dots, e), \quad \mathbf{E}_n D_{E_h, n} \simeq G(B_{h, n}), \quad D_{E_h, n}[\mathbf{E}_n^{-1}] \simeq \mathbf{E}_n D_{E_h, n}. \quad (\text{MCF31})$$

Its multiplication fills every missing coordinate of an active input with a supported zero while leaving all ordinary coefficients unchanged; every output has a supported summand using any chosen active input coordinate. Hence $\mathbf{E}_n^2 = \mathbf{E}_n$. A unital map making this idempotent invertible sends it to 1 and identifies c with $\mathbf{E}_n c$, proving the localization universal property. If $f = \sum f_i t^i$ and $J(f) = \{i : f_i \neq 0\}$, its fibre over $f^\bullet$ consists of one original vector for each nonempty mask A with $J(f) \subseteq A \subseteq I_n$. The fibre over τ is the all-absent vector alone. In particular a supported zero has $2^n - 1$ distinct independent-mask preimages. These exact fibres remain attached; the synchronized observation does not replace (MCF28).

Finally the original arithmetic coefficients have an exact retract

$$\iota_n : E_h \longrightarrow B_{h, n}, \quad f \mapsto f, \quad r_n : B_{h, n} \longrightarrow E_h, \quad r_n(b) = \frac{1}{n} \text{Tr}_{B_{h, n}/E_h}(M_b), \quad r_n \iota_n = 1. \quad (\text{MCF32})$$

In the basis $1, t, \dots, t^{n-1}$, multiplication by t^i has no diagonal entries for $1 \leq i < n$, while multiplication by b_0 has n diagonal entries b_0 . Thus $r_n(\sum b_i t^i) = b_0$ and its kernel is exactly $\bigoplus_{i=1}^{n-1} E_h t^i$. Writing Ψ_{q_0} for (MCF30) on coefficients, it fixes ι_n and preserves r_n . The coefficientwise extension of $A_{a, h}$ commutes with it and intertwines both arrows in (MCF32). On a coefficient eigenspace of eigenvalue ω , define the combined operator $T_{a, q_0} = A_{a, h} \Psi_{q_0} = \Psi_{q_0} A_{a, h}$ on $B_{h, n}$. Its actual full local action is

$$\omega a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (\text{MCF33})$$

Every ω is a root of unity by the proved finite order, and the original arithmetic action is recovered on the constant-coefficient retract. This completes the attachment of independent mixed coefficient support to the original theta source, its combined leg diagram, and its actual arithmetic operators. All masks, source diagonals, full units, nilpotents, and comparison fibres have their displayed maps.

1.7 The independent outer source label, including empty coefficient support

The original outer source label is (W, S) , with $W \subseteq V$ a complex linear D -invariant subspace and $\emptyset \neq S \subseteq \{+, -\}$. It is independent of the coefficient mask $A \subseteq I_n$. Denote the set of those subspaces W by $\mathcal{W}_D$. Its exact coefficient diagram is

$$\begin{aligned} \mathcal{J}_n^{\text{out}} &= \{(W, S; A) : W \in \mathcal{W}_D, \emptyset \neq S \subseteq \{+, -\}, A \subseteq I_n\}, \\ C_{W, S; A}^0 &= \left(\bigoplus_{+ \in S} W \oplus \bigoplus_{- \in S} \widehat{W} \right)^A, \quad C_{W, S; A}^1 = \mathcal{B}_U^A, \\ d_{W, S; A}(\phi_i, \psi_i)_i &= (\Theta \phi_i - \Theta \widehat{\psi}_i)_i. \end{aligned} \quad (\text{MCF34})$$

The minus leg is exactly $\widehat{W}$; it is not replaced by W unless the two spaces are equal. Its action is $1 - D$, preserving that space by the Fourier identity. The target restriction $\mathcal{B}_U \subseteq \mathcal{B}$ is an injective cochain map to the original outer complex, since $\Theta W \subseteq \Theta V \subseteq \mathcal{B}_U$. For $W \subseteq W'$, $S \subseteq S'$ and $A \subseteq A'$, the transition includes the old source subspaces and coordinates and inserts the typed zeros in new legs and coefficient slots. The degree-one transition is zero insertion. These are cochain maps, by substitution in the displayed differential, and compose literally.

At $A = \emptyset$ this is a zero complex *at its retained outer label* (W, S) . Its single coefficient value has empty coefficient support; the outer leg mask is still S . In the represented carrier one retains all triples $(W, S; A, x)$ and separately adjoins a global external point τ_{out} . In particular

$$(W, S; \emptyset, 0) \neq \tau_{\text{out}}, \quad (W, S; A, 0) \neq (W, S; A', 0) \quad \text{if } A \neq A'. \quad (\text{MCF35})$$

These are explicit distinct elements of the disjoint labelled carrier; they do not assert a nonzero amplitude in a zero complex. Taking a colimit over coefficient masks would identify these zero insertions with their images in the terminal full face. That colimit is not part of this construction.

Define $Q_{W,U} = \mathcal{B}_U / \Theta W$ and $\eta_W : V/W \rightarrow Q_{W,U}$ by $[v] \mapsto [\Theta v]_W$. The complete arithmetic quotient and kernel are

$$0 \longrightarrow (V/W)^A \xrightarrow{\eta_W^A} Q_{W,U}^A \xrightarrow{\pi_W^A} (T_U Q)^A \longrightarrow 0, \quad (\text{MCF36})$$

$$H^0(C_{W,S;A}) = \begin{cases} W^A, & S = \{+, -\}, \\ 0, & |S| = 1, \end{cases} \quad H^1(C_{W,S;A}) = Q_{W,U}^A.$$

For the kernel sequence, injectivity of Θ proves that $\Theta v \in \Theta W$ exactly when $v \in W$. A class of $\mathcal{B}_U / \Theta W$ maps to zero modulo ΘV exactly when it is represented by Θv for some $v \in V$. Surjectivity follows from the definition of $\mathcal{B}_U$. Applying this proof independently in the coordinates proves the exact sequence. For the cohomology calculation, a joint slot has the reversible coordinates (MCF14) with $u, v \in W$, since $\psi \in \widehat{W}$. Its differential is Θu and its diagonal is W . A single-leg differential is injective with image ΘW . This proves every case, including the zero coefficient vector spaces at empty A . For $W \subseteq W'$ the transition on degree one has kernel exactly $(W'/W)^A$ under the same original map $[v] \mapsto [\Theta v]_W$. The outer-label kernel is thus fully retained.

The exact relation to the slot-local diagram is an enlarged index

$$(W, S; A, (T_i)_{i \in A}), \quad \emptyset \neq T_i \subseteq S. \quad (\text{MCF37})$$

Its source uses W on the present plus legs and $\widehat{W}$ on the present minus legs, according to T_i , and its target is $\mathcal{B}_U^A$. Every differential is the original signed theta differential in that coordinate. The outer diagram (MCF34) embeds by $T_i = S$ for every $i \in A$, with the identity on its coefficient spaces. This embedding remains injective on the index when A is empty because W and S are retained separately. Forgetting W, S and including $W, \widehat{W}$ into V gives the explicit cochain comparison to (MCF12); its target map is the identity on $\mathcal{B}_U^A$. Its induced map on H^1 is precisely π_W^A in (MCF36), and its induced map on H^0 is the diagonal inclusion $W^{\{i:T_i=\{+,-\}\}} \subseteq V^{\{i:T_i=\{+,-\}\}}$. For fixed $W = V$, forgetting only the outer S has exactly three labelled preimages over the empty slot-local mask, two when the union of the nonempty T_i is a single leg, and one when that union contains both legs. Indeed these are exactly the nonempty subsets $S \subseteq \{+, -\}$ containing $\bigcup_i T_i$. This computes the lost outer information of that specified forgetting map; its empty coefficient value is not identified with global absence in (MCF35).

There is a useful original-outer-source retract, which also checks the empty coefficient endpoint without changing the label. At a fixed (W, S) , write $C_{W,S}^U$ for its one-coefficient original complex with central term $\mathcal{B}_U$, and take the constant coefficient map

$$c_0 : C_{W,S;A} \longrightarrow C_{W,S}^U, \quad c_0(x) = \begin{cases} x_0, & 0 \in A, \\ 0, & 0 \notin A, \end{cases} \quad i_0 : C_{W,S}^U \longrightarrow C_{W,S;\{0\}}, \quad x \longmapsto (x)_0. \quad (\text{MCF38})$$

Here i_0 includes the original outer complex as the coefficient-face vertex $A = \{0\}$ in the full represented diagram. This notation does not assert an inclusion into every other face. Each c_0 is a cochain map and commutes with all coefficient zero insertions, since a newly inserted coordinate has zero amplitude. On the represented carrier it sends $(W, S; \emptyset, 0)$ to the supported outer zero at (W, S) . On global τ_{out} the pointed extension has value τ_{out} . Its composite with the selected i_0 is the identity on the original outer coefficient amplitude and label. The coefficient Frobenius fixes index zero and its sign is one, so $c_0 \Psi_{q_0} = c_0$. This proves an actual outer-label-preserving comparison, rather than identifying the original outer label with a slot-local mask.

The finite arithmetic section at a proper outer source is $\sigma_{h,W}(u) = [s_h u]_W \in Q_{W,U}$. It is injective because $\pi_W \sigma_{h,W} = \sigma_h$ and σ_h is injective. Its exact retained defects are

$$\begin{aligned} D\sigma_{h,W} - \sigma_{h,W} A_h &= \eta_W([\phi_* \ell_h(\cdot)]_W), \\ \overline{\mathcal{R}}_a^{W \rightarrow \mathcal{R}_a W} \sigma_{h,W} - \sigma_{h,\mathcal{R}_a W} A_{a,h} &= \eta_{\mathcal{R}_a W}([\Psi_{a,h}(\cdot)]_{\mathcal{R}_a W}). \end{aligned} \quad (\text{MCF39})$$

The first is (MCF5) in the quotient by ΘW and the second is (MCF24) in the quotient by $\Theta \mathcal{R}_a W$. Dilation transports the outer label to $\mathcal{R}_a W$, as required by (MCF19); it need not fix an arbitrary D -invariant W . On the minus source it maps $\widehat{W}$ to $a\mathcal{R}_{1/a}\widehat{W} = \widehat{\mathcal{R}_a W}$. This proves that every arrow in (MCF39) has the stated domain and codomain. The original primitives belong to V ; their possibly nonzero classes in V/W or $V/\mathcal{R}_a W$ are retained in (MCF36). No primitive in V has been declared to be a boundary in a smaller source W . For the full outer source $W = V$, both kernel classes are zero and (MCF39) recovers the proved full-source intertwining.

Apply (MCF39) coordinatewise on every A , keeping all outer and slot-local labels. Packet inclusion, zero insertion, and coefficient Frobenius commute with these defects by (MCF27), linearity and the permutation (MCF30). In particular the complete retained data form the commuting represented diagram

$$\{C_{W,S;A}\}_{W,S,A} \longrightarrow \{Q_{W,U}^A[-1]\}_{W,S,A} \xrightarrow{\{\pi_W^A\}} \{(T_U Q)^A[-1]\}_{W,S,A}, \quad \ker \pi_W^A = \eta_W^A((V/W)^A), \quad (\text{MCF40})$$

with the enriched slot-local faces and their inclusion and forgetting maps specified in (MCF37). The first arrow is the actual cochain map to $Q_{W,U}^A[-1]$ given by zero in degree zero and the original quotient in degree one. Its exact kernel complex is also retained:

$$K_{W,S;A} = [C_{W,S;A}^0 \xrightarrow{d} (\Theta W)^A], \quad 0 \longrightarrow K_{W,S;A} \longrightarrow C_{W,S;A} \longrightarrow Q_{W,U}^A[-1] \longrightarrow 0. \quad (\text{MCF40a})$$

Degreewise kernel computation proves this short exact sequence. The single-leg differential onto ΘW is an isomorphism, and the joint differential in (MCF14) has kernel the full W diagonal. Thus $K_{W,S;A}$ retains exactly the degree-zero diagonal specified in (MCF36), including its zero-insertion and source-inclusion maps. The original outer source, every coefficient mask including the empty one, every present source leg, and the exact arithmetic comparison kernel are simultaneously specified by this diagram.

1.8 Exact enlargement from offcritical support to all nontrivial packets

The offcritical region U throughout the preceding construction remains unchanged. A packet containing critical-line zeros has a larger original source domain, which we now attach by its exact inclusions. Put

$$\begin{aligned} \Omega &= \{s \in \mathbb{C} : 0 < \Re s < 1\}, \quad L = \Omega \setminus U = \{s : \Re s = \tfrac{1}{2}\}, \\ T_X Q &= \bigcup_{p \in \mathcal{P}_X} \ker(p(D) : Q \rightarrow Q), \quad \mathcal{B}_X = q^{-1}(T_X Q), \quad C_X = [V \xrightarrow{\Theta} \mathcal{B}_X] \quad (X = U, \Omega, L), \\ \mathcal{P}_X &= \{p \text{ monic} : \text{every root of } p \text{ belongs to } X\}, \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}]. \end{aligned} \quad (\text{MCF41})$$

The polynomial 1 is in every $\mathcal{P}_X$. The notation C_L denotes an algebraic source complex; no openness of the line L or sheaf restriction to an open L is required. The same polynomial range identities (MCF3) hold for every root in the open strip Ω , including its critical line.

The exact source inclusions, with their original degree-zero maps, are

$$C_U[A] \xrightarrow{(1, \text{inc})} C_\Omega[A] \xrightarrow{(1, \text{inc})} C_+[A]. \quad (\text{MCF41a})$$

Indeed $\mathcal{P}_U \subseteq \mathcal{P}_\Omega$ gives $T_U Q \subseteq T_\Omega Q$ and then $\mathcal{B}_U \subseteq \mathcal{B}_\Omega \subseteq \mathcal{B}$. All differentials are the same original Θ and the degree-zero maps are the identity on V^A , proving both cochain identities. Injectivity holds in each degree. Cohomology in degree zero is zero by injectivity of Θ , and the induced degree-one sequence is the literal sequence of submodules $(T_U Q)^A \hookrightarrow (T_\Omega Q)^A \hookrightarrow Q^A$.

Its full quotient has the exact arithmetic meaning

$$T_\Omega Q = T_U Q \oplus T_L Q, \quad T_\Omega Q / T_U Q \xrightarrow{\sim} T_L Q. \quad (\text{MCF41b})$$

Here is a direct proof without presuming a list of existing zeros. For $u \in T_\Omega Q$, choose an annihilator p in $\mathcal{P}_\Omega$ and factor it as $p = p_U p_L$, retaining every multiplicity, with roots of p_U in U and roots of p_L in L . Either factor can be 1. The factors have disjoint roots, so there are polynomials a, b with $ap_U + bp_L = 1$. Then

$$u_U = b(D)p_L(D)u, \quad u_L = a(D)p_U(D)u, \quad u = u_U + u_L, \quad (\text{MCF41c})$$

and multiplication shows $p_U(D)u_U = p_L(D)u_L = 0$. If a vector is killed by a polynomial rooted in U and by a polynomial rooted in L , the same Bezout identity makes that vector zero. Thus the two submodules intersect in zero and the decomposition is unique, independent of the chosen annihilator and Bezout polynomials. This proves (MCF41b), including its inverse: include $T_L Q$ in $T_\Omega Q$ and take its class modulo $T_U Q$.

There is an exact original Mellin description of this quotient as well:

$$T_X Q \xrightarrow{\sim} \bigoplus_{\substack{\rho \in X \\ g(\rho)=0}} \mathcal{O}_\rho / (g), \quad [F] \mapsto ([\mathcal{M}F]_\rho)_\rho, \quad X = U, L, \Omega. \quad (\text{MCF41d})$$

To check the map, write $p(D)F = \Theta\phi$. Then $p(s)\mathcal{M}F(s) = g(s)H_\phi(s)$, so the Mellin class is zero at every zero of g outside the finite root set of p . Replacing F by a theta boundary changes its Mellin transform by gH and does not change a local class. For injectivity, suppose all local classes at roots of p vanish. At a root with g nonzero, g is a local unit; at a root of g , the vanishing class says that $\mathcal{M}F$ is a multiple of g . In either case write $\mathcal{M}F = gB$ locally. Cancellation in the local analytic domain gives $H_\phi = pB$, so $j_p H_\phi = 0$ at every required order. By (MCF3), $\phi = p(D)\psi$ with $\psi \in V$. Injectivity of $p(D)$ on $\mathcal{B}$ then yields $F = \Theta\psi$.

For surjectivity, take a finitely supported family of full local jets α_ρ at actual zeros in X . For its finite support Z put $h = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$ with the complete actual orders and choose the unique polynomial P of degree less than $\deg h$ with $j_h P = j_h (g/h)^{-1} \alpha$. The complete unit exists at each specified centre and CRT supplies that polynomial. The original Euler range identity gives

$$F_\alpha = S_h^\mathcal{B} \Theta(P(D)\phi_*), \quad \mathcal{M}F_\alpha = (g/h)P, \quad h(D)F_\alpha \in \Theta V. \quad (\text{MCF41e})$$

These equations give the specified full jets and zero classes at all other zeros of g . The empty family uses the stage $h = 1$ and the zero class. Changing P by hR changes F_α by the original boundary $\Theta(R(D)\phi_*)$. This proves both inverse identities in (MCF41d), retaining the entire g/h , every coordinate $s - \rho$, and every multiplicity. Under this isomorphism the decomposition (MCF41b) is exactly zero insertion and restriction to the disjoint offcritical and critical-line sets of full local factors.

Degree-wise quotients of the actual complexes now give

$$\begin{aligned} 0 &\rightarrow C_U[A] \rightarrow C_\Omega[A] \rightarrow (T_\Omega Q / T_U Q)^A[-1] \rightarrow 0, \\ 0 &\rightarrow C_\Omega[A] \rightarrow C_+[A] \rightarrow (Q / T_\Omega Q)^A[-1] \rightarrow 0. \end{aligned} \quad (\text{MCF41f})$$

Their degree-zero quotients vanish because both inclusions are the identity on V^A . For the first degree-one quotient, the map $\mathcal{B}_\Omega/\mathcal{B}_U \rightarrow T_\Omega Q/T_U Q$ induced by q is onto, and has zero kernel because $qF \in T_U Q$ is exactly $F \in \mathcal{B}_U$. The second quotient is proved by the identical calculation with $\mathcal{B}/\mathcal{B}_\Omega$ and $Q/T_\Omega Q$. These prove the displayed short exact sequences on the original source domains. In particular a nonzero critical local factor survives in the first quotient; it has not been discarded by working with the original offcritical subcomplex.

For every actual nontrivial full packet h in Ω , including a critical pair or a mixture of critical and offcritical centres, the formulas (MCF4)–(MCF5) and (MCF22)–(MCF27) have their unchanged values and their source now lies in $\mathcal{B}_\Omega$ because $h(D)s_h u \in \Theta V$. Their proofs used (MCF3) in the open strip and the full local unit; they did not use $\Re \rho \neq 1/2$. Thus they apply to that precise larger domain. Their restriction to packets rooted in U agrees literally with the earlier formulas through (MCF41a). Every packet system includes $h = 1$ with $E_1 = 0$; no nonempty offcritical packet is assumed to make the enlarged construction or its embeddings.

Finally retain every coefficient and outer label in this enlargement. Replacing only the central term of (MCF34) by $\mathcal{B}_X^A$ defines $C_{W,S,A}^X$ and $Q_{W,X} = \mathcal{B}_X/\Theta W$. The maps from $X = U$ to $X = \Omega$ are the identity on the original source legs and the inclusion on the central term, and their degree-one quotient is again $(T_\Omega Q/T_U Q)^A$:

$$0 \longrightarrow Q_{W,U}^A \longrightarrow Q_{W,\Omega}^A \longrightarrow (T_\Omega Q/T_U Q)^A \longrightarrow 0. \quad (\text{MCF41g})$$

Indeed $\Theta W \subseteq \mathcal{B}_U$; quotienting the two nested central subspaces by that same subspace proves injectivity and the stated quotient. The proper-source kernel $(V/W)^A$ of (MCF36) remains attached on each side. Empty coefficient support still retains (W, S) as in (MCF35). The enlarged maps commute with zero insertion, the specified slot-local comparisons, dilation and its proper-source transport, and coefficient Frobenius, since their source maps are identities and their central maps are the original coordinatewise inclusions. This attaches all actual nontrivial packets while preserving the original offcritical construction and its full comparison quotient.

1.9 Literal comparison with the earlier finite source and its cocycle

The earlier source (TC1) writes the spectral polynomial coordinate as t and the same residue representative as $R_Z(u)$; the coordinate isomorphism sends t to s , with $h_Z(t)$ sent to $h(s)$, every power $(t-\rho)^r$ sent to $(s-\rho)^r$, and the full unit unchanged. Consequently $R_Z(u)$ is $P_h(u)$ under this specified isomorphism. For the actual source functions, not only their quotient classes,

$$R_{\text{ref}} = s_h : E_h \longrightarrow \mathcal{B}_X, \quad X = U \text{ or } \Omega \text{ as the packet requires.} \quad (\text{MCF42})$$

Indeed applying $h(D)$ to either representative gives $\Theta(P_h(u)(D)\phi_*)$. The difference lies in $\mathcal{B}$ and is killed by the injective $h(D)$, hence vanishes. This proves the asserted equality with the original units and domains.

Put $t = \log a$, retaining its orientation when $a < 1$, and define

$$c_{t,h}(u) = \int_0^t \mathcal{R}_{e^{t-r}\phi_*} \ell_h(e^{rA_h} u) dr : E_h \longrightarrow V. \quad (\text{MCF43})$$

The integral converges in the original Schwartz topology. In fact on the compact interval joining 0 and t , every derivative of $\mathcal{R}_{e^{t-r}\phi_*}$ is a continuously varying rescaled derivative of the same Schwartz function, with the scaling parameter in a compact subset of $(0, \infty)$. For every pair N, k , its seminorm $\sup_x (1 + |x|)^N |\partial_x^k(\cdot)|$ is uniformly bounded there. The finite matrix e^{rA_h} is smooth and bounded there too. Riemann sums therefore converge in every seminorm in the complete Schwartz space; parity and the two zero moments pass to the limit by continuity, so the limit is in V .

Differentiate the actual $\mathcal{B}$ -valued function $\mathcal{R}_{e^{t-r}s_h} e^{rA_h} u$. Its derivative is $-\mathcal{R}_{e^{t-r}}(Ds_h - s_h A_h) e^{rA_h} u$. The same seminorm argument on $\mathcal{B}$, followed by (MCF5), allows the fundamental

theorem of calculus. It gives

$$\Theta_{c_{\log a, h}} = \mathcal{R}_a s_h - s_h A_{a, h}, \quad \Psi_{a, h} = c_{\log a, h}. \quad (\text{MCF44})$$

The second equality follows by comparing with (MCF24) and canceling the injective map $\Theta : V \rightarrow \mathcal{B}$. Thus the previous oriented integral and the Euler-inverse primitive are the same map. Their packet and coefficient-face naturality are literal, by (MCF27) and coordinatewise integration. The joint primitive is $(c/2, -\widehat{c}/2)$; the difference between the two single-leg primitives $(c, 0)$ and $(0, -\widehat{c})$ is the full cycle $(c, \widehat{c})$. No diagonal is discarded by this comparison.

The earlier dual two-leg complex is obtained by applying algebraic $\text{Hom}_{\mathbb{C}}(-, \mathbb{C})$, so the map induced by $C_U \rightarrow C_+$ is restriction $\mathcal{B}^\vee \rightarrow \mathcal{B}_U^\vee$ in degree -1 and identity on both $V^\vee$ source coordinates in degree zero. Its differential, in the same order of legs, is

$$\ell \mapsto (\ell\Theta, -\ell\Theta\mathcal{F}), \quad \mathcal{F}\phi = \widehat{\phi}. \quad (\text{MCF45})$$

Because $\Theta V \subseteq \mathcal{B}_U$, restriction followed by this differential equals the differential followed by the identity. Thus this is an actual cochain map. It induces restriction $Q^\vee \rightarrow (T_U Q)^\vee$ on degree -1 cohomology. At a joint source the two diagonal parameters $v \mapsto (v, \mathcal{F}v)$ and $w \mapsto (\mathcal{F}w, w)$ agree under $w = \mathcal{F}v$. Define $\mathcal{F}^\vee : V^\vee \rightarrow V^\vee$ by $\mathcal{F}^\vee(\ell) = \ell \circ \mathcal{F}$; the maps $\pi_v, \pi_w : V^\vee \oplus V^\vee \rightarrow V^\vee$ are

$$\pi_v(\alpha, \beta) = \alpha + \beta\mathcal{F}, \quad \pi_w(\alpha, \beta) = \alpha\mathcal{F} + \beta, \quad \pi_v = \mathcal{F}^\vee \circ \pi_w. \quad (\text{MCF46})$$

These identities follow by evaluation on the displayed diagonals and $\mathcal{F}^2 = 1$. The original actions intertwine because $\mathcal{F}\mathcal{R}_a = a\mathcal{R}_{1/a}\mathcal{F}$. For a proper source W , dualizing $W \hookrightarrow V$ gives restriction of diagonal functionals to W . For coefficient faces, dualizing zero insertion gives coordinate restriction. These are algebraic dual statements; no continuous splitting for arbitrary nonclosed W is asserted.

1.10 The shifted coefficient quotient on the original theta complexes

Fix one of the actual full packets h in Ω and a member ρ with $m = m_\rho \geq 2$. The calculation is indexed by precisely those members; it requires no assertion that the indexing set is nonempty. The coordinate isomorphism from the x presentation in (MW23) to this s presentation sends x to s and $e_\rho(x - \rho)^r$ to $e_\rho(s - \rho)^r$, with coefficient one and inverse $s \mapsto x$. It carries the full defining polynomial, its CRT idempotents, and every retained jet order. Use the original local subspace $E_\rho = e_\rho E_h$, $J_\rho = (s - \rho)^2 E_\rho$, and $C_\rho = E_\rho / J_\rho$ of (MW23). Write σ_ρ for the restriction of the original injective $\sigma_h : E_h \rightarrow Q$. Define actual subspaces of the original central source by

$$\mathcal{B}_\rho = q^{-1}(\sigma_\rho E_\rho), \quad \mathcal{B}_{J_\rho} = q^{-1}(\sigma_\rho J_\rho), \quad \Theta V \subseteq \mathcal{B}_{J_\rho} \subseteq \mathcal{B}_\rho \subseteq \mathcal{B}_\Omega. \quad (\text{MCF47})$$

The last inclusion follows because every displayed quotient class lies in the $h(D)$ -torsion, and h is rooted in Ω . The literal direct-sum equalities are

$$\mathcal{B}_\rho = \Theta V \oplus s_h E_\rho, \quad \mathcal{B}_{J_\rho} = \Theta V \oplus s_h J_\rho. \quad (\text{MCF48})$$

For the first, if $qF = \sigma_\rho x$, then $F - s_h x \in \Theta V$; injectivity of σ_ρ makes this x unique. If $\Theta v + s_h x = 0$, quotienting gives $x = 0$, then injectivity of Θ gives $v = 0$. The same argument with $x \in J_\rho$ proves the second equality. Every representative here is the original $s_h x = P_h(x)(D)F_h$ with the full unit $j_h(g/h)^{-1}$, including its components at all selected centres.

There is an actual short exact sequence of source complexes

$$0 \longrightarrow [V \xrightarrow{\Theta} \mathcal{B}_{J_\rho}] \xrightarrow{(1, \text{inc})} [V \xrightarrow{\Theta} \mathcal{B}_\rho] \longrightarrow C_\rho[-1] \longrightarrow 0. \quad (\text{MCF49})$$

The final map is zero in degree zero. In degree one it sends F to $[x]$, where the unique $x \in E_\rho$ satisfies $qF = \sigma_\rho x$. It is linear, vanishes on ΘV , is onto because $s_h x$ is a representative of every

$[x]$, and has kernel exactly $\mathcal{B}_{J_\rho}$. These assertions prove both the cochain identity and degreewise exactness. Its inverse on the degree-one quotient sends $[x]$ to $[s_h x] \in \mathcal{B}_\rho / \mathcal{B}_{J_\rho}$; replacing x by $x + j$, $j \in J_\rho$, changes its representative by the actual element $s_h j \in \mathcal{B}_{J_\rho}$. Both inverse identities follow from (MCF48). Thus the length-two quotient used in (MW24)–(MW27) and (MRE34)–(MRE42) has this explicit original-source realization.

The arithmetic actions in this exact sequence also have their original types. Both E_ρ and J_ρ are invariant under A_h , $N_\rho = A_h - \rho I$ on E_ρ , and each $A_{a,h}$. The full-source intertwining (MCF5), (MCF21) makes D , $D - \rho$, and $\mathcal{R}_a$ preserve the corresponding spaces in (MCF47). Their maps on degree zero are respectively D , $D - \rho$, and $\mathcal{R}_a$ on V , and their maps on degree one use the same original operators on functions. Since each commutes with Θ , they are cochain maps. Under the final map in (MCF49) they give exactly the quotient $\overline{A}_h$, $\overline{N}_\rho$, and $\overline{A}_{a,h}$ of (MW25), with every surviving coefficient retained. The full higher powers remain in the original source operators and in the subcomplex on the left.

The induced filtration is realized before taking cohomology. For every integer j , set

$$\begin{aligned} \mathcal{B}_{\rho,j} &= q^{-1}(\sigma_\rho(M_j E_\rho)), & \mathcal{B}_{J_\rho,j} &= q^{-1}(\sigma_\rho(J_\rho \cap M_j E_\rho)) = \mathcal{B}_{J_\rho} \cap \mathcal{B}_{\rho,j}, \\ 0 \longrightarrow [V \xrightarrow{\Theta} \mathcal{B}_{J_\rho,j}] &\longrightarrow [V \xrightarrow{\Theta} \mathcal{B}_{\rho,j}] \longrightarrow \overline{M}_j C_\rho[-1] \longrightarrow 0. \end{aligned} \quad (\text{MCF50})$$

The intersection equality uses injectivity of σ_ρ . For the exact sequence, the last map of (MCF49) sends $\mathcal{B}_{\rho,j}$ onto $\pi_\rho(M_j E_\rho)$, because the representatives $s_h x$ exist for every $x \in M_j E_\rho$. Its kernel is precisely the displayed intersection. The degree-zero map remains identity and the last differential is zero, proving every case. Consequently the quotient filtration is exactly (MW24), with degrees $m - 1$ and $m - 3$, hence with its retained shift $m - 2$ relative to the displayed length-two centered filtration.

For all sufficiently negative j , the two central terms in (MCF50) equal ΘV . Their source complexes then remain $[V \xrightarrow{\Theta} \Theta V]$, with its actual invertible differential; no assertion that these source complexes are literally zero is made. They have zero cohomology because Θ is injective and onto its displayed image. At every step the original D preserves $\mathcal{B}_{\rho,j}$, and $D - \rho$ sends it into $\mathcal{B}_{\rho,j-2}$ since its quotient action is precisely N_ρ . This proves the filtered source-map types as well as the induced filtered cohomology identities.

For each proper original $W \subseteq V$, replace the degree-zero term in (MCF49)–(MCF50) by the outer source $\bigoplus_{+ \in S} W \oplus \bigoplus_{- \in S} \widehat{W}$ with the original signed theta differential. Use that identical source in the left and middle complexes. The same proof of degreewise exactness applies, and in degree one their individual cohomology groups have the exact sequences

$$\begin{aligned} 0 \longrightarrow V/W &\longrightarrow \mathcal{B}_{\rho,j}/\Theta W \longrightarrow M_j E_\rho \longrightarrow 0, \\ 0 \longrightarrow V/W &\longrightarrow \mathcal{B}_{J_\rho,j}/\Theta W \longrightarrow J_\rho \cap M_j E_\rho \longrightarrow 0. \end{aligned} \quad (\text{MCF51})$$

In both lines $[v] \mapsto [\Theta v]_W$ is the original kernel map, and the last map is $\sigma_\rho^{-1} q$ on its indicated image. Injectivity, surjectivity, and the kernel computation are the same as in (D11). Thus the low-filtration degree-one cohomology is V/W at a proper source, while the joint degree-zero diagonal is W ; both remain present. For D use $DW \subseteq W'$ and the specified source codomain W' as in (D14). For $D - \rho$ use $(D - \rho)W \subseteq W'$; its minus source operator is $1 - D - \rho$, because $\mathcal{F}(D - \rho) = (1 - D - \rho)\mathcal{F}$. The choice $W' = W + DW$ admits both maps. Dilation uses $\mathcal{R}_a W$. Apply the same proof independently at every coefficient mask, including its retained empty outer label. Zero insertion and coefficient Frobenius commute with the displayed quotient maps by linearity and their original signed coordinate permutation. This transports the strict shifted coefficient quotient to the original mixed source system without removing its actual comparison kernels.

2 Reconstruction with changes of support index

The web formalization constructs reverse reconstruction, its fibrewise quotients, and an inverse on each actual semimodule. This section gives the explicit categorical comparison, including changes

of support index, used to connect those objects to the arithmetic chart. No assumption of injective transition maps or zero bottom fibre is made.

Fix a commutative ring R and its split semiring $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$. A diagram object consists of a join-semilattice L with bottom and R -modules V_l , with linear maps $\rho_{lk} : V_l \rightarrow V_k$ for $l \leq k$, satisfying identity and composition. A morphism from (L, V) to (L', W) is a pair (α, f) , where $\alpha : L \rightarrow L'$ preserves bottom and joins with R -linear fibre maps

$$f_l : V_l \rightarrow W_{\alpha(l)}, \quad f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l. \quad (D1)$$

Composition is $(\beta, g)(\alpha, f) = (\beta\alpha, (g_{\alpha(l)} f_l)_l)$. These are the literal carriers and maps of the diagram category.

Theorem 2.1 (Full support-index reconstruction). *The diagram category just defined is equivalent to the category of $G(R)$ -semimodules and their $G(R)$ -linear maps. The forward functor has carrier*

$$\mathcal{T}(L, V) = \coprod_{l \in L} V_l$$

and operations

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k} x + \rho_{k, l \vee k} y), \quad \tau(l, x) = (\perp, 0), \quad r^\bullet(l, x) = (l, rx). \quad (D2)$$

Its global zero is $(\perp, 0)$. On morphisms it is $\mathcal{T}(\alpha, f)(l, x) = (\alpha(l), f_l x)$.

The inverse sends a semimodule M to

$$\begin{aligned} L_M &= eM, \quad l \vee k = l + k, \quad \perp = 0_M, \\ V_l &= \{m \in M : em = l\}, \quad \rho_{lk}(m) = m + k \quad (l + k = k). \end{aligned} \quad (D3)$$

The natural comparison maps are

$$\mathcal{T}(L_M, V) \longrightarrow M, \quad (l, m) \mapsto m, \quad m \mapsto (em, m), \quad (D4)$$

and the label-and-fibre isomorphism

$$L \longrightarrow e\mathcal{T}(L, V), \quad l \mapsto (l, 0), \quad x \in V_l \mapsto (l, x). \quad (D5)$$

Proof. For (D2), threefold addition transports both parenthesizations to $l \vee k \vee j$. Linearity and composition of the ρ 's make both amplitudes $\rho_l x + \rho_k y + \rho_j z$ in that same fibre. The bottom zero and commutativity follow likewise. For supported scalars the module laws follow from those of each V_l . Their addition law at a vector in fibre l is the ordinary R -module addition in V_l ; in particular $0_R^\bullet$ gives $(l, 0)$. The τ cases use the global bottom zero. Distributivity over vectors follows from linearity of each transition. These checks prove every semimodule axiom. Equation (D1) proves that $\mathcal{T}(\alpha, f)$ respects addition; its scalar and zero laws are immediate. Identity and composition hold on each pair.

For the reverse direction, $e^2 = e$, $e + e = e$, and $1 + e = 1$ in $G(R)$. Thus every $l \in eM$ is additively idempotent; eM is closed under addition, which supplies a semilattice with the specified bottom. For $m \in V_l$,

$$m + l = (1 + e)m = m, \quad m + (-1_R)^\bullet m = em = l.$$

The inherited addition in V_l , with zero l and inverse $(-1_R)^\bullet m$, is therefore an abelian group. For any $r \in R$, $e(r^\bullet m) = em = l$, since $er^\bullet = e$. Its scalar action satisfies every R -module law inherited from the semimodule, and $0_R m = em = l$ is its local zero. This proves that V_l is the specified module, without cancellation assumptions on all of M .

If $l + k = k$, then $e(m + k) = l + k = k$, so ρ_{lk} has the correct codomain. Its zero law is $l + k = k$; its additivity uses $k + k = k$. Also $r^\bullet k = k$, proving R -linearity. Identity follows from $m + l = m$, and

composition from $k + j = j$. A split-linear map $F : M \rightarrow N$ takes eM to eN , with $\alpha(l) = F(l)$. It preserves joins and bottom, restricts to R -linear maps on each fibre, and $F(m + k) = F(m) + F(k)$ proves (D1). This defines the inverse functor.

Both arrows in (D4) are inverse as functions. Addition on the reconstructed carrier has amplitude

$$(m + (l \vee k)) + (n + (l \vee k)) = m + n + (l \vee k) = m + n,$$

because $e(m + n) = l \vee k$. Thus (D4) is additive; its scalar identities follow from (D2), including τ . It is a split-linear isomorphism. In the reconstructed total, $e(l, x) = (l, 0)$, so (D5) is an isomorphism of semilattices and of the original fibres. Its transition map is exactly $(l, x) + (k, 0) = (k, \rho_{lk}x)$, recovering the given ρ . For a morphism (α, f) , (D5) takes $(l, 0)$ to $(\alpha(l), 0)$; for F , (D4) sends a representative to $F(m)$. These equalities prove naturality of both inverse comparisons, including support-index changes. Hence they are a categorical equivalence. $\square$

2.1 The internal quotient and the actual comparison kernel

Let $B_l \subseteq V_l$ be transition-stable submodules. The quotient diagram has fibres V_l/B_l with their induced transitions. The map

$$q : \mathcal{T}(L, V) \rightarrow \mathcal{T}(L, V/B), \quad (l, x) \mapsto (l, [x]) \quad (\text{D6})$$

is onto and split-linear. It is the coequalizer of the two maps $\mathcal{T}(L, B) \rightrightarrows \mathcal{T}(L, V)$ given by inclusion and $(l, b) \mapsto (l, 0)$.

To prove the universal property against any split-linear target N , let f equalize these maps. If $x - y \in B_l$,

$$f(l, x) = f(l, y) + f(l, x - y) = f(l, y) + f(l, 0) = f(l, y).$$

The last equality is the image of $(l, y) + (l, 0) = (l, y)$; no subtraction or cancellation in N was used. Thus f descends as a function through (D6). Lifting representatives proves additivity, scalar laws and uniqueness. Different labels remain distinct under (D6). In particular a boundary is carried to its original support-labelled zero.

For a cochain map $f : C \rightarrow D$ of R -module complexes at a fixed degree, put $Z_C = \ker d_C$, $B_C = \text{im } d_C^{\text{prev}}$ inside Z_C , and $K_{\text{rep}} = \{z \in Z_C : f(z) \in B_D\}$. Then the map

$$K_{\text{rep}}/B_C \xrightarrow{\sim} \ker H(f), \quad [z] \mapsto [z] \quad (\text{D7})$$

is a linear isomorphism. It is well-defined since f maps boundaries to boundaries. The prequotient map $K_{\text{rep}} \rightarrow \ker H(f)$, $z \mapsto [z]$, has kernel exactly B_C ; every kernel class has a cycle representative, proving surjectivity. If $fa = bf$ for chain endomorphisms, $aK_{\text{rep}} \subseteq K_{\text{rep}}$ and $aB_C \subseteq B_C$; (D7) therefore intertwines the actual induced actions. If f is a morphism of diagrams of complexes, these constructions commute with the transitions and reconstruct by (D2). For the equivariant assertion across supports, a, b are diagram endomorphisms as well. The equation $d^2 = 0$ then becomes the map $(l, x) \mapsto (l, 0)$ from its original source degree to its original target degree. It is different from constant absence when $l \neq \perp$, and (D6)–(D7) are the exact comparison maps.

For completeness, the full semimodule category retains two precisely related kernel constructions. For $F = \mathcal{T}(\alpha, f) : M = \mathcal{T}(L, V) \rightarrow N = \mathcal{T}(L', W)$, put $L_0 = \alpha^{-1}(\perp')$. Then

$$\begin{aligned} \ker_{G(R)} F &= \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}), \\ F^{-1}(eN) &= \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN. \end{aligned} \quad (\text{D8})$$

The inclusion of the first into the second is induced by $L_0 \hookrightarrow L$. Indeed L_0 contains bottom and is closed under joins. Naturality in (D1) takes each fibre kernel into the next one. The formula $F(l, x) = (\alpha(l), f_l x)$ equals the global zero exactly when $\alpha(l) = \perp'$ and $f_l x = 0$; it belongs to eN exactly when $f_l x = 0$. These pointwise descriptions prove both universal properties: any split-linear

map satisfying the respective condition has its unique factorization through the displayed subsemi-module. Thus reconstruction of the all-label fibre kernels in (D7) gives the second object. If α is the identity, the categorical kernel is its bottom fibre, with the inclusion in (D8) retaining the exact comparison.

2.2 The earlier reconstruction applied to every original source level

We now evaluate (D2), (D6), and (D7) on the original theta source, including the finite polynomial source levels used in (TC2) and (TB.24)–(TB.26). Let $\mathcal{W}$ contain every complex linear subspace W of the original V , ordered by inclusion; no closure or invariance under D is imposed for this cochain construction. Fix $n \geq 1$ and $I_n = \{0, \dots, n-1\}$. Its nonbottom labels are

$$\lambda = (W, S; A), \quad W \in \mathcal{W}, \quad \emptyset \neq S \subseteq \{+, -\}, \quad A \subseteq I_n. \quad (\text{D9})$$

Adjoin one distinct bottom $\perp$, with zero fibre. The join of two nonbottom labels is $(W + W', S \cup S'; A \cup A')$. The bottom is strictly below even $(0, S; \emptyset)$. In degrees zero and one put

$$C_{W,S;A}^{X,0} = \left(\bigoplus_{+ \in S} W \oplus \bigoplus_{- \in S} \widehat{W} \right)^A, \quad C_{W,S;A}^{X,1} = \mathcal{B}_X^A, \quad (\text{D10})$$

$$d_{W,S;A}(\phi_i, \psi_i)_i = (\Theta\phi_i - \Theta\widehat{\psi}_i)_i.$$

Here $X = U$ or Ω as in (MCF41); missing legs use typed zeros. The transition includes old subspaces and coordinates and inserts typed zeros in newly present legs or coefficients. Fourier linearity gives $\widehat{W} \subseteq \widehat{W + W'}$; hence all transitions have their stated domains. Substitution into the differential proves the cochain identity, and inclusions compose literally. Thus (D2) applies in each degree to this actual diagram. At every nonbottom label with $A = \emptyset$ the amplitude fibre is zero, but its element is $(W, S; \emptyset, 0)$, while the global zero is $(\perp, 0)$. Applying (D2) proves they remain distinct and gives their exact joins. No coefficient-mask colimit is taken.

For each W , put $Q_{W,X} = \mathcal{B}_X / \Theta W$. The full cohomology and the comparison to the full source are

$$H^0(C_{W,S;A}^X) = \begin{cases} W^A, & S = \{+, -\}, \\ 0, & |S| = 1, \end{cases} \quad H^1(C_{W,S;A}^X) = Q_{W,X}^A, \quad (\text{D11})$$

$$0 \longrightarrow (V/W)^A \xrightarrow{\eta_W^A} Q_{W,X}^A \xrightarrow{\pi_W^A} (T_X Q)^A \longrightarrow 0, \quad \eta_W[v] = [\Theta v]_W.$$

Indeed the source differential has image $(\Theta W)^A$. A single-leg differential is injective. At a joint slot the inverse coordinates $u = \phi - \widehat{\psi}$, $v = (\phi + \widehat{\psi})/2$, $\phi = v + u/2$, $\psi = \widehat{v} - \widehat{u}/2$ identify its kernel with $v \in W$. For the last sequence, injectivity of Θ makes η_W injective; a class killed by π_W is represented by Θv for some $v \in V$, and every element of $T_X Q$ has a representative in $\mathcal{B}_X$. This proves exactness, coordinatewise also for empty A . The coequalizer (D6) uses precisely $(\Theta W)^A$, and (D7) identifies the kernel of the full-source comparison with $(V/W)^A$ through this same η_W^A . This is the all-label fibre kernel of (D8); with unchanged support label, the global-zero categorical kernel contains only its bottom fibre. Both maps are retained.

For $W \subseteq W'$ the source transition therefore induces

$$0 \longrightarrow (W'/W)^A \xrightarrow{[v] \mapsto [\Theta v]_W} Q_{W,X}^A \longrightarrow Q_{W',X}^A \longrightarrow 0. \quad (\text{D12})$$

The proof uses the same kernel computation with W' in place of V . Thus source refinement has its actual nonzero comparison kernel before any full-source quotient is applied.

The critical-strip enlargement also has its original typed map:

$$0 \longrightarrow C_{W,S;A}^U \longrightarrow C_{W,S;A}^\Omega \longrightarrow (T_L Q)^A[-1] \longrightarrow 0, \quad L = \{\Re s = 1/2\}. \quad (\text{D13})$$

The first map is identity on source legs and inclusion on the central term. Its degree-zero quotient is zero. Its degree-one quotient is $\mathcal{B}_\Omega^A/\mathcal{B}_U^A$, since the same ΘW lies in both. The quotient map q identifies this space with $(T_\Omega Q/T_U Q)^A$. For any annihilator $p = p_U p_L$ with its factors rooted in the disjoint sets U, L , choose $ap_U + bp_L = 1$. The maps $b(D)p_L(D)$ and $a(D)p_U(D)$ split each killed class into its U and L parts. Their intersection is zero by the same Bezout identity; uniqueness makes the splitting independent of the annihilator. Thus the quotient is exactly $(T_L Q)^A$, with every multiplicity retained. All three complexes keep the same outer and coefficient labels.

The original arithmetic operator has the following additional type on this larger source diagram. If $DW \subseteq W'$, then

$$D_{W,W'} : C_{W,S;A}^X \longrightarrow C_{W',S;A}^X, \quad (\phi_i, \psi_i)_i \longmapsto (D\phi_i, (1-D)\psi_i)_i, \quad (F_i)_i \longmapsto (DF_i)_i. \quad (\text{D14})$$

Fourier gives $(1-D)\widehat{W} = \widehat{DW} \subseteq \widehat{W}'$, and $\Theta D = D\Theta$ proves the cochain identity. Moreover $D\mathcal{B}_X \subseteq \mathcal{B}_X$ because D commutes with each polynomial annihilator of a quotient class. Thus every displayed map has its original domain. For D -invariant W , take $W' = W$ to recover precisely (MCF34). For the actual finite source levels $W_m = \text{span}\{\phi_*, D\phi_*, \dots, D^m\phi_*\}$, one instead has $DW_m \subseteq W_{m+1}$ and must use (D14) with that codomain. Linear independence of these vectors follows from $H_{P(D)\phi_*} = P$, so $\dim(W_{m+1}/W_m) = 1$ exactly. Dilation likewise gives a cochain map from W to $\mathcal{R}_a W$, using $\mathcal{R}_a$ on the plus source and central term and $a\mathcal{R}_{1/a}$ on the minus source. The Fourier identity proves its stated minus codomain $\widehat{\mathcal{R}_a W}$.

3 Exact source-metric optimization and aggregate arithmetic control

This section continues the actual theta complexes in the September 12 chain-descent, global-comparison, Deligne-control, and orthogonal-boundary notes. It proves an aggregate bound on the distinguished orthogonal representatives and computes exactly the optimization over all finite polynomial theta corrections. Every matrix below is recorded in the original polynomial coordinates of the packet. Auxiliary comparison maps and metrics are written explicitly.

3.1 The fixed arithmetic maps and their boundary

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx). \end{aligned}$$

The Hilbert inner product is conjugate-linear in its first variable. The original transform conventions are

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The original source vector and its iterates are

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta \phi_j, \quad \mathcal{M}f_j(s) = s^j g(s), \quad g = 2\xi.$$

Fix a nonempty finite set Z of actual nontrivial zeros with their full multiplicities m_ρ , and retain

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z, \quad E = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

The coordinate basis of E is $1, t, \dots, t^{d-1}$. Let j_h denote the full Taylor-jet/Chinese-remainder map, let $\varepsilon_Z = j_h(h_Z/g)$, and write $R_Z(u)$ for the unique polynomial of degree less than d representing $\varepsilon_Z u$. The finite source construction gives

$$\begin{aligned} R_{\text{ref}} u &= S_{h_Z}^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad JF = j_h \mathcal{M}F, \quad \sigma_Z = qR_{\text{ref}}, \\ JR_{\text{ref}} &= 1_E, \quad J\Theta = 0, \quad DR_{\text{ref}} - R_{\text{ref}}A = f_0 \ell_Z, \quad \ell_Z(u) = [t^{d-1}]R_Z(u). \end{aligned} \quad (\text{TC1})$$

Here $S_{\rho}^{\mathcal{B}} F(x) = x^{-\rho} \int_x^{\infty} F(y) y^{\rho-1} dy$, on its zero-Mellin-moment domain, and its iterated inverse defines $S_{h_Z}^{\mathcal{B}}$. Thus all local units in g/h_Z and the factor two in $g = 2\xi$ remain present.

For $L \geq 0$ define the actual finite source maps

$$\Phi_L : \mathbb{C}^{L+1} \longrightarrow V, \quad c \longmapsto \sum_{j=0}^L c_j \phi_j, \quad F_L = \Theta \Phi_L : \mathbb{C}^{L+1} \longrightarrow \mathcal{B}.$$

Every allowed representative in the family considered here has the form

$$R = R_{\text{ref}} + F_L C, \quad C : E \longrightarrow \mathbb{C}^{L+1}. \quad (\text{TC2})$$

In particular $qR = \sigma_Z$ and $JR = 1_E$. Let $I_L, S_L : \mathbb{C}^{L+1} \rightarrow \mathbb{C}^{L+2}$ send the standard coordinate vector e_j respectively to e_j, e_{j+1} . Directly,

$$DR - RA = \Theta K_R, \quad K_R = \phi_* \ell_Z + \Phi_{L+1}(S_L C - I_L C A) : E \longrightarrow V. \quad (\text{TC3})$$

This formula keeps the entire source boundary. Its class in Q is zero through $q\Theta = 0$.

Put $G_R = R^* R$, a positive definite Hermitian matrix in the fixed packet coordinates. For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

because $[x \overline{F(x)} H(x)]_0^{\infty} = 0$. Substitution of (TC3) therefore proves the exact identity

$$W_R := A^* G_R + G_R A - G_R = -(R^* \Theta K_R + (\Theta K_R)^* R). \quad (\text{TC4})$$

The adjoints here are adjoints of maps from finite-dimensional coefficient spaces to $\mathcal{H}$; no Hilbert extension of Θ is presumed.

3.2 A source-density calculation with its topology retained

We give the density argument used below, so the realization theorem has its full analytic input available. The map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dy), \quad (\mathcal{U}F)(y) = e^{y/2} F(e^y)$$

is unitary by $x = e^y$. Fourier transformation with kernel e^{ity} and measure $dt/(2\pi)$ sends $\mathcal{U}f_j$ to $(1/2 + it)^j g(1/2 + it)$.

The Gaussian theta series defining $f_0(e^z)$ converges holomorphically when $|\text{Im } z| < \pi/4$. On every closed narrower strip, $\text{Re}(e^{2z}) \geq c_a e^{2\text{Re } z}$, with $c_a = \cos(2a) > 0$; its polynomial Gaussian summands consequently have superexponential decay as $\text{Re } z \rightarrow +\infty$. The source ϕ_* is Fourier-fixed: the Gaussian is Fourier-fixed and the transform sends $D(D-1)$ to $(1-D)(-D) = D(D-1)$. Poisson summation and its two zero moments give $f_0(x) = x^{-1} f_0(1/x)$. Thus the holomorphic function $k(z) = e^{z/2} f_0(e^z)$ obeys $k(z) = k(-z)$, first on the real axis and then throughout the strip. The same decay holds at the negative end. Moving the Fourier contour to $\text{Im } z = a$ sign t has vanishing vertical edges and proves, for every $0 < a < \pi/4$,

$$|g(1/2 + it)| \leq C_a e^{-a|t|}. \quad (\text{TC5})$$

Let $d\mu(t) = |g(1/2 + it)|^2 dt/(2\pi)$. This finite positive measure has a positive exponential moment. If $v \in L^2(\mu)$ is orthogonal to every polynomial, the finite complex measure $d\nu = \bar{v} d\mu$ has a smaller positive exponential moment, by Cauchy–Schwarz. Its Fourier transform is holomorphic on a strip and all derivatives at zero vanish, by the polynomial orthogonality. Its power series and analytic continuation make it identically zero. Uniqueness of the Fourier transform of finite measures gives $\nu = 0$, hence $v = 0$ in $L^2(\mu)$. Polynomials are therefore dense in $L^2(\mu)$. Multiplication by $g(1/2 + it)$ is a unitary map from $L^2(\mu)$ onto $L^2(dt/(2\pi))$: its norm equality is the definition of μ ; surjectivity follows by division off the measure-zero zero set of the nonzero entire function g . It follows that

$$\overline{\bigcup_{L \geq 0} F_L \mathbb{C}^{L+1}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{TC6})$$

This is a statement in the displayed Hilbert topology. The identities $JR = 1$ below continue to hold in the original test-function space.

3.3 Every positive packet metric has an exact finite source lift

Define

$$H_m = F_m^* F_m, \quad C_m = F_m^* R_{\text{ref}}, \quad B_m = -H_m^{-1} C_m, \quad \Pi_m = F_m H_m^{-1} F_m^*, \quad R_m = R_{\text{ref}} + F_m B_m. \quad (\text{TC7})$$

The columns of F_m are linearly independent, because a relation has Mellin transform $g(s)P(s) = 0$ and hence $P = 0$. Thus $H_m > 0$, Π_m is the orthogonal projection onto the stated finite theta space, and

$$F_m^* R_m = 0, \quad G_m := R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \quad (\text{TC8})$$

Its strict positivity follows from $JR_m = 1$. Equation (TC6) proves $\Pi_m \rightarrow 1$ strongly. Since E is finite dimensional, $R_m \rightarrow 0$ in $\text{Hom}(E, \mathcal{H})$, and $G_m \rightarrow 0$ in matrix norm.

Exact fixed-level image. At every specified level m , the entire image of the source-Gram map is

$$\{(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) : C : E \rightarrow \mathbb{C}^{m+1}\} = \{G_m + \Delta : \Delta \succeq 0, \text{rank } \Delta \leq m+1\}. \quad (\text{TC8a})$$

Indeed completing the square, using $F_m^* R_m = 0$, gives

$$(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) = G_m + (C - B_m)^* H_m (C - B_m). \quad (\text{TC8b})$$

The extra term is positive semidefinite and has the stated rank bound. Conversely, a positive semidefinite Δ of rank at most $m+1$ has a factorization $\Delta = T^* T$ with $T : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$: diagonalize the finite Hermitian matrix Δ , retain its nonnegative eigenvalues, and insert their square roots into $m+1$ rows. Taking $C = B_m + H_m^{-1/2} T$ proves the converse. The auxiliary factorization records a coefficient map; it changes none of the original packet data.

Exact realization theorem. For every specified positive definite Hermitian matrix G in the original polynomial coordinates of E , there exist a finite L and a coefficient map $C : E \rightarrow \mathbb{C}^{L+1}$ for which

$$R = R_{\text{ref}} + F_L C, \quad JR = 1, \quad qR = \sigma_Z, \quad R^* R = G. \quad (\text{TC9})$$

The construction is as follows. Choose $m \geq d-1$ with $G_m \prec G$, set $L = m$, and let $I_{d-1,m} : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$ pad a vector with zeros. The principal matrix $I_{d-1,m}^* H_m I_{d-1,m}$ is exactly H_{d-1} . Put

$$\Delta = G - G_m > 0, \quad C = B_m + I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}. \quad (\text{TC10})$$

Both square roots are the uniquely specified positive Hermitian roots of finite positive matrices on the displayed coefficient spaces, using their fixed standard Hermitian forms and the fixed polynomial-coordinate Hermitian form on E . The resulting map remains expressed in the original packet coordinates. Indeed

$$R = R_m + F_m I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}, \quad R_m^* F_m = 0,$$

so its Gram is

$$R^* R = G_m + \Delta^{1/2} H_{d-1}^{-1/2} H_{d-1} H_{d-1}^{-1/2} \Delta^{1/2} = G.$$

All added columns are finite complex linear combinations of the original $D^j \phi_*$ followed by Θ . This proves (TC9) inside the declared source family, with a finite coefficient degree.

The extension class has the literal type $\text{Ext}_{\mathbb{C}[t]}^1(E, V) \simeq V/h_Z(D)V$; its arithmetic coordinate ε_Z has type E . The exact map between these types is

$$\iota_h : E \longrightarrow V/h_Z(D)V, \quad [p(t)] \longmapsto [p(D)\phi_*]. \quad (\text{TC10a})$$

It is well-defined and $\mathbb{C}[t]$ -linear because replacing p by $p + h_Z a$ adds $h_Z(D)a(D)\phi_*$. It is an isomorphism, with the explicit inverse

$$\kappa_h : V/h_Z(D)V \longrightarrow E, \quad [\psi] \longmapsto j_h \left(\frac{\mathcal{M}\psi}{\mathcal{M}\phi_*} \right). \quad (\text{TC10b})$$

The denominator is

$$\mathcal{M}\phi_*(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2),$$

which is holomorphic and nonzero near every $\rho \in Z$. Every $\psi \in V$ has $\psi(x) = O(x^2)$ at zero, so its Mellin transform is holomorphic for $\text{Re } s > -2$. Mellin integration by parts gives $\mathcal{M}(p(D)\psi) = p(s)\mathcal{M}\psi$ near Z ; this proves well-definedness of κ_h and $\kappa_h \iota_h = 1_E$. For the other inverse law, let $\psi \in V$ have vanishing Mellin jets along h_Z . For the first factor $t - \rho$, its source inverse on $x > 0$ is

$$S_\rho \psi(x) = x^{-\rho} \int_x^\infty \psi(y) y^{\rho-1} dy = - \int_0^1 \psi(tx) t^{\rho-1} dt.$$

The second expression and $0 < \text{Re } \rho < 1$ prove smooth even extension to zero, with value zero; differentiating in every order is justified by the integrable factor $t^{\text{Re } \rho - 1}$. The first expression gives Schwartz decay at infinity. Integration of $(D - \rho)S_\rho \psi = \psi$ on the full real line gives $(1 - \rho) \int S_\rho \psi = \int \psi = 0$, so $S_\rho \psi \in V$. Its Mellin transform is $\mathcal{M}\psi(s)/(s - \rho)$, so the remaining zero jets permit iteration through every factor of h_Z . Consequently $\ker(j_h \mathcal{M}|_V) = h_Z(D)V$. Apply this to $\psi - p(D)\phi_*$, where $p = j_h(\mathcal{M}\psi/\mathcal{M}\phi_*)$, to obtain $\iota_h \kappa_h = 1$. Thus the class of the actual arithmetic extension is precisely $\iota_h(\varepsilon_Z)$, which is nonzero. For every source change (TC2), this class is unchanged. Explicitly,

$$h_Z(D)R(1) = \Theta(R_Z(1)(D)\phi_* + h_Z(D)\Phi_L C(1)),$$

and the second term is zero in $V/h_Z(D)V$. There is still no strictly D -equivariant section: an intertwiner $T : E \rightarrow \mathcal{B}$ has image killed by $h_Z(D)$, whereas $h_Z(D)$ is injective on $\mathcal{B}$. To see this injectivity, $(D - \rho)F = 0$ implies $F(x) = cx^{-\rho}$; the bounds defining $\mathcal{B}$ force $c = 0$, and one iterates over the factors of h_Z . The exact map relating any section to the original arithmetic action is (TC3), with its retained nonzero extension class.

3.4 Sharp control with all nilpotent jets present

For a positive definite G , set

$$W_A(G) = A^*G + GA - G, \quad \epsilon_A(G) = \sup_{u \neq 0} \frac{|u^* W_A(G) u|}{u^* G u}, \quad \eta_Z = \max_{\rho \in Z} |2 \text{Re } \rho - 1|. \quad (\text{TC11})$$

For every eigenvector $Au = \rho u$,

$$u^* W_A(G) u = (2 \operatorname{Re} \rho - 1) u^* G u. \quad (\text{TC12})$$

Consequently $\epsilon_A(G) \geq \eta_Z$, for every positive G . We now construct metrics arbitrarily close to that bound without removing a single nilpotent jet.

Fix $\epsilon > \eta_Z$, and fix any $G_0 > 0$, both in the original coordinates. Define

$$\begin{aligned} F(t) &= e^{-t} e^{tA^*} G_0 e^{tA}, \\ G_\epsilon &= \int_{-\infty}^{\infty} e^{-\epsilon|t|} F(t) dt, \\ W_A(G_\epsilon) &= \epsilon \int_{-\infty}^{\infty} \operatorname{sign}(t) e^{-\epsilon|t|} F(t) dt. \end{aligned} \quad (\text{TC13})$$

The last identity is a conclusion, proved below. The factor e^{-t} records the original weight-one character; the action is still $e^{tA} = U_Z(e^t)$.

Let $e_\rho \in E$ be the exact Chinese-remainder idempotent and write $P_\rho = M_{e_\rho}$ and $N_\rho = (A - \rho I)P_\rho$. On $P_\rho E$, $N_\rho^{m_\rho} = 0$. Directly,

$$e^{tA} = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho. \quad (\text{TC14})$$

Thus every matrix entry in the integral is a finite sum of polynomial-exponential functions. The real part of each exponent $\bar{\rho} + \sigma - 1$ has absolute value at most η_Z . This proves absolute convergence of (TC13), including its derivative, and proves the endpoint limits needed for integration by parts. In particular $G_\epsilon > 0$. Since $F'(t) = A^* F(t) + F(t)A - F(t)$, integration by parts on the negative and positive half-lines gives (TC13). The two values at $t = 0$ cancel exactly. Both half-line integrals are strictly positive matrices, so

$$-\epsilon G_\epsilon \prec W_A(G_\epsilon) \prec \epsilon G_\epsilon. \quad (\text{TC15})$$

For completeness the construction requires no unevaluated analytic integral once the packet data and G_0 are specified. Put

$$I_k(z, \epsilon) = k! \left((\epsilon - z)^{-k-1} + (-1)^k (\epsilon + z)^{-k-1} \right). \quad (\text{TC16})$$

The substitutions $t \mapsto -t$ on the negative half-line and $\int_0^\infty t^k e^{-at} dt = k! a^{-k-1}$ show that

$$\begin{aligned} G_\epsilon &= \sum_{\rho, \sigma \in Z} \sum_{i=0}^{m_\rho-1} \sum_{j=0}^{m_\sigma-1} \frac{I_{i+j}(\bar{\rho} + \sigma - 1, \epsilon)}{i! j!} \\ &\quad \cdot (N_\rho^i P_\rho)^* G_0 (N_\sigma^j P_\sigma). \end{aligned} \quad (\text{TC17})$$

All local Jordan powers and every cross term remain in this formula. Combining (TC12), (TC15), and the actual realization (TC9) proves

$$\boxed{\inf_{G>0} \epsilon_A(G) = \inf_{\substack{L \geq 0 \\ C: E \rightarrow \mathbb{C}^{L+1}}} \epsilon_A((R_{\text{ref}} + F_L C)^* (R_{\text{ref}} + F_L C)) = \eta_Z.} \quad (\text{TC18})$$

The equality concerns all finite polynomial corrections. It does not assert that the distinguished orthogonal sequence (TC7) attains the same infimum.

Exact attainment criterion. The infimum in (TC18) is achieved precisely when every eigenvalue on either boundary $|2 \operatorname{Re} \rho - 1| = \eta_Z$ is semisimple. For $E = \mathbb{C}[t]/h_Z$, this means $m_\rho = 1$ for those zeros.

To prove necessity, a bound $\epsilon_A(G) \leq \eta_Z$ integrates to

$$\|e^{tA}u\|_G \leq e^{(1+\eta_Z)t/2}\|u\|_G \quad (t \geq 0),$$

and applying the other inequality backward gives

$$\|e^{-tA}u\|_G \leq e^{(-1+\eta_Z)t/2}\|u\|_G \quad (t \geq 0).$$

On a generalized eigenspace on the upper boundary the first estimate bounds $e^{t(A-\rho I)}$ uniformly for $t \geq 0$, up to a unit-modulus scalar. A nonzero nilpotent block has a vector whose exponential contains a nonzero highest power of t , hence is unbounded. The lower boundary is treated by the backward estimate. This proves necessity.

For sufficiency, use the unchanged CRT direct-sum map $u \mapsto (P_\rho u)_\rho$. Make its summands mutually orthogonal by taking the pullback sum of chosen positive metrics on them. On a semisimple boundary summand, $A = \rho I$, so the control form is exactly $(2 \operatorname{Re} \rho - 1)$ times that summand's metric. For each interior summand choose a number strictly between $|2 \operatorname{Re} \rho - 1|$ and η_Z , and use (TC13) on that summand. Its control is strictly smaller than η_Z . The sum therefore has the required bound. When $\eta_Z = 0$ there are only boundary summands, and the same construction gives zero control exactly. The source realization (TC9) then supplies an actual finite corrected representative for this metric. Its boundary (TC3) persists; zero metric control only states that its Hermitian cross-pairing in (TC4) is zero.

3.5 The exact tensor cost of unrestricted metrics

For the original ordered tensor space $E^{\otimes n}$, retain

$$A_n = \sum_{j=1}^n I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(n-j)}, \quad W_{A_n, n}(G) = A_n^* G + G A_n - nG. \quad (\text{TC19})$$

Its eigenvalues, with algebraic multiplicity, are all ordered sums $\rho_1 + \dots + \rho_n$. This follows by tensoring complete invariant flags, or directly by the CRT decomposition and the nilpotent sum on each ordered product block. Hence

$$\max_{\lambda \in \operatorname{Spec} A_n} |2 \operatorname{Re} \lambda - n| = n\eta_Z. \quad (\text{TC20})$$

The upper bound is the triangle inequality; equality uses the ordered repeated tuple $(\rho, \dots, \rho)$ for a zero attaining η_Z . Repeating (TC12)–(TC17) with the weight factor e^{-nt} proves the exact value

$$\boxed{\inf_{G>0} \sup_{v \neq 0} \frac{|v^* W_{A_n, n}(G) v|}{v^* G v} = n\eta_Z.} \quad (\text{TC21})$$

The nilpotent sum on a product block is retained throughout this argument. Thus an unrestricted choice of a product-image metric, even a non-product metric, has a precisely calculated linear spectral cost. Establishing sublinear excess for the actual arithmetic image requires mathematical information about its spectrum or a further comparison which constrains that spectrum. Equations (TC18) and (TC21) are calculations of the optimization; they supply no missing estimate by choosing representatives.

3.6 A stronger bound on the actual orthogonal sequence

Return to the specific R_m in (TC7), and take Z stable under $\rho \mapsto 1 - \bar{\rho}$, including multiplicities. Write

$$b_m = e_m^* B_m : E \rightarrow \mathbb{C}, \quad z_m = R_m^* f_{m+1} : \mathbb{C} \rightarrow E.$$

The source boundary of R_m lies in the span of $f_0, \dots, f_{m+1}$. Orthogonality to the first $m+1$ vectors gives directly from (TC3)–(TC4)

$$W_m = A^* G_m + G_m A - G_m = -z_m b_m - b_m^* z_m^*, \quad \text{rank } W_m \leq 2. \quad (\text{TC22})$$

Reflection stability gives

$$\text{Tr}(G_m^{-1} W_m) = 2 \text{Re Tr } A - d = 0. \quad (\text{TC23})$$

Let $\epsilon_m = \epsilon_A(G_m)$. The G_m -self-adjoint map $G_m^{-1} W_m$ therefore has at most two nonzero eigenvalues, namely $\epsilon_m, -\epsilon_m$; if the rank is zero both are zero.

Aggregate displacement theorem. For every finite m in this original orthogonal source sequence,

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \max(2 \text{Re } \rho - 1, 0) = \sum_{\rho \in Z} m_\rho \left| \text{Re } \rho - \frac{1}{2} \right|. \quad (\text{TC24})$$

Here the second equality uses the given reflection involution and its preservation of multiplicities.

For a full proof of the first inequality, take a complete A -invariant flag, keeping all generalized eigenvectors. Choose a G_m -orthonormal frame adapted to that flag and denote its explicit coordinate map by $T : \mathbb{C}^d \rightarrow E$. It satisfies $T^* G_m T = I$; it is only a map used in the proof, while the recorded A, G_m, W_m remain in the original coordinates. The matrix $T^{-1} A T$ is upper triangular with diagonal entries the zeros ρ , repeated m_ρ times. Therefore the Hermitian matrix $H = T^* W_m T$ has diagonal entries $\delta_j = 2 \text{Re } \rho_j - 1$. By (TC22)–(TC23) its positive part H_+ has trace ϵ_m . Let P be the orthogonal coordinate projection onto the indices with $\delta_j > 0$. As $0 \leq P \leq I$ and $H \leq H_+$,

$$\sum_{\delta_j > 0} \delta_j = \text{Tr}(PH) \leq \text{Tr}(PH_+) \leq \text{Tr}(H_+) = \epsilon_m. \quad (\text{TC25})$$

Each trace inequality follows by inserting the positive square root of P or H_+ , so no sign is inferred from a product of noncommuting matrices. This proves (TC24).

More generally, whenever the positive rank of a control form is r_+ , the same argument gives $\sum_j \max(\delta_j, 0) \leq r_+ \epsilon_A(G)$. In the present source construction $r_+ \leq 1$, which is why the aggregate bound has no dimension loss.

The source's exact scalar certificate is retained:

$$u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m, \quad \epsilon_m^2 = u_m v_m - (\text{Im } c_m)^2. \quad (\text{TC26})$$

Indeed the two nonzero eigenvalues are those of $-\begin{pmatrix} c_m & v_m \\ u_m & c_m \end{pmatrix}$, and (TC23) gives $\text{Re } c_m = 0$. If $a_{m+1} = (1 - \Pi_m) f_{m+1}$ and $\eta_{m+1} = \|a_{m+1}\|^2$, the exact rank-one Gram update gives

$$G_{m+1} = G_m - \eta_{m+1}^{-1} z_m z_m^*, \quad u_m = \eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right). \quad (\text{TC27})$$

The determinant identity is the rank-one determinant lemma, obtained by choosing a basis whose first vector spans $G_m^{-1} z_m$, or by expanding the determinant multilinearly. Thus a fully arithmetic version of (TC24) is

$$\eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right) v_m - (\text{Im } c_m)^2 \geq \left(\sum_{\rho \in Z} m_\rho \left| \text{Re } \rho - \frac{1}{2} \right| \right)^2. \quad (\text{TC28})$$

Every entry on the left is an actual theta Gram or cross-Gram entry with the original source representative. This identifies a stronger target for the arithmetic moment estimates than a bound for a single eigenvector.

3.7 Original supported maps at every independent source and coefficient face

The passage from a finite source correction C to its representative, Gram, and boundary retains the earlier exact full-source formula

$$C \mapsto R_{\text{ref}} + \Theta \Phi_L C \mapsto (qR, JR, R^*R, K_R) = (\sigma_Z, 1_E, G_R, K_R). \quad (\text{TC29})$$

Here is its complete map before the full-source quotient. Set $T = \Phi_L C : E \rightarrow V$, so $R = s_h + \Theta T$ by (MCF42). At an arbitrary original source subspace $W \subseteq V$, define $\sigma_{R,W}(u) = [Ru]_W \in Q_{W,\Omega}$. Then

$$\sigma_{R,W} - \sigma_{h,W} = \eta_W([T(\cdot)]_W), \quad \pi_W \sigma_{R,W} = \sigma_Z. \quad (\text{TC29a})$$

These follow by subtracting the actual functions and using (D11). Consequently a permitted full-source correction is invisible in Q but retains precisely its class in V/W at a smaller source. The map $\sigma_{R,W}$ is injective, since its composite with π_W is the injective σ_Z .

If $DW \subseteq W'$, the generator comparison has its exact type

$$\overline{D}_{W,W'} \sigma_{R,W} - \sigma_{R,W'} A = \eta_{W'}([K_R(\cdot)]_{W'}), \quad K_R = \phi_* \ell_Z + DT - TA. \quad (\text{TC29b})$$

Indeed (TC3) is the source-function equality $DR - RA = \Theta K_R$; taking its class modulo $\Theta W'$ proves the formula. The left action exists by (D14), so no invariant source subspace is presumed here. Its kernel class vanishes exactly when $K_R(u) \in W'$ for the vector in question, by the injectivity of $\eta_{W'}$.

For every $a > 0$ the full corrected dilation primitive is

$$\begin{aligned} c_{R,a} &= \Psi_{a,h} + \mathcal{R}_a T - TA_{a,h} : E \rightarrow V, \\ \mathcal{R}_a R - RA_{a,h} &= \Theta c_{R,a}, \\ \overline{\mathcal{R}}_a^{W \rightarrow \mathcal{R}_a W} \sigma_{R,W} - \sigma_{R,\mathcal{R}_a W} A_{a,h} &= \eta_{\mathcal{R}_a W}([c_{R,a}]_{\mathcal{R}_a W}). \end{aligned} \quad (\text{TC29c})$$

The first two equalities follow by substituting $R = s_h + \Theta T$ into (MCF24) and using $\mathcal{R}_a \Theta = \Theta \mathcal{R}_a$. The last follows by quotienting by $\Theta \mathcal{R}_a W$. Expanding the first expression and applying (MCF25) gives

$$c_{R,ab} = \mathcal{R}_a c_{R,b} + c_{R,a} A_{b,h}. \quad (\text{TC29d})$$

In the expansion, the terms $\mathcal{R}_a T A_{b,h}$ cancel with their negative, leaving $\mathcal{R}_{ab} T - TA_{ab,h}$ and the original primitive $\Psi_{ab,h}$. This proves the law with its complete source action and ordered arithmetic product.

At a label $(W, S; \mathfrak{a})$, with $\mathfrak{a} \subseteq I_n$, apply all these maps independently to each coordinate $u_i \in E$ for $i \in \mathfrak{a}$. The proper-source residual is then in $(V/W')^{\mathfrak{a}}$ or $(V/\mathcal{R}_a W)^{\mathfrak{a}}$, respectively. When a primitive belongs to the declared source, its two single-leg lifts are $(K_R u, 0)$ and $(0, -\widehat{K_R u})$; their difference is the entire diagonal $(K_R u, \widehat{K_R u})$. If it does not belong, its class in (D11) is retained instead. The labels $W, S, \mathfrak{a}$ are unchanged by evaluating a zero amplitude, and an empty $\mathfrak{a}$ retains the outer label; external absence is still $(\perp, 0)$. Zero insertion commutes with the linear formulas, as does the coefficient Frobenius using the same sign in source and target. Every corrected metric and boundary formula (TC4)–(TC28) therefore uses the same original functions as before, while (TC29a)–(TC29d) record their full mixed-source comparison.

The new conclusions are the exact fixed-level Gram image (TC8a), the finite source realization (TC9), the sharp source-metric infimum with its attainment criterion (TC18), the exact tensor infimum (TC21), and the aggregate arithmetic bound (TC24)–(TC28). The metric realization leaves the extension unit, the residue/Jacobian pairing, the quotient classes and the generator unchanged. The original distinguished orthogonal metrics remain a smaller, arithmetically specified family. Their relative estimates, including all shrinking denominators in (TC26), are the next concrete arithmetic calculation.

4 The actual theta minimum as a confluent arithmetic kernel

This section continues the orthogonal-boundary construction of the 12 September 2026 tau workbench, through PR 13. Its new calculation identifies the permitted finite source corrections with a constrained polynomial minimum, evaluates that minimum by a complete jet kernel, and proves the exact two-vector boundary formula for that kernel. The arithmetic packet, Euler operator, local units, and multiplicities are retained throughout.

4.1 Original functions, maps, coordinates, and measures

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x.$$

Let $\mathcal{B}$ be the space of smooth functions on $\mathbb{R}_{>0}$ such that $\sup_{x>0} x^b |D^k F(x)| < \infty$ for every integer b and every $k \geq 0$. The original maps and function are

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*, \quad g(s) = 2\xi(s), \quad \mathcal{M}f_0 = g.$$

Write $\phi_j = D^j \phi_*$ and $f_j = D^j f_0 = \Theta\phi_j$. The Hilbert observation is $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$; the first argument of every inner product is conjugate-linear. The actual Mellin-line map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dt/(2\pi)), \quad \mathcal{U}F(t) = \mathcal{M}F(1/2 + it)$$

is unitary. Indeed $x = e^y$ gives the unitary map $F \mapsto e^{y/2} F(e^y)$ to $L^2(dy)$, and the displayed Mellin transform is its Fourier transform with the positive exponential e^{ity} . Plancherel has measure $dt/(2\pi)$ in this convention. Integration by parts on $\mathcal{B}$ gives $\mathcal{U}DF = (1/2 + it)\mathcal{U}F$.

Fix a nonempty finite packet Z of actual nontrivial zeros of g , including the full multiplicity m_ρ of each selected zero. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[s]/(h), \quad A = M_s : E \longrightarrow E.$$

The basis of E remains $1, s, \dots, s^{d-1}$. Let j_h denote polynomial remainder together with its equivalent complete Taylor-jet description. Set

$$v(s) = \frac{g(s)}{h(s)}, \quad \varepsilon = (j_h v)^{-1} \in E^\times, \quad U_\varepsilon = M_\varepsilon.$$

The function v is entire and has nonzero value at each selected centre after the corresponding cancellation. The matrix U_ε retains all derivatives of g/h , including the factor 2 in $g = 2\xi$. It commutes with A . For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree below d representing εu . Write $R_{\text{ref}} : E \rightarrow \mathcal{B}$ for the source note's fixed finite section R_0 . This alias records its identity and avoids colliding with the $m = 0$ corrected member of the sequence R_m . The fixed section is characterized by

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \quad J_Z R_{\text{ref}} = I_E, \quad J_Z \Theta = 0. \quad (\text{TB.1})$$

These are the actual finite section and jets of the tau construction; the proofs below do not replace them with a finite model.

Introduce the two specified measures on the same Mellin line

$$d\mu(t) = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad d\nu_Z(t) = |v(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{TB.2})$$

Both have every polynomial moment. For μ this follows from the Gaussian-series Mellin decay proved in PR 13; division by the fixed polynomial h preserves that decay outside a compact set, and the removable values of v are bounded on the compact set. The densities are positive except on discrete sets, because g and v are nonzero entire functions. Consequently the squared norm of every nonzero polynomial in either measure is strictly positive. No positive arithmetic residue or Weil pairing has been inserted in (TB.2).

4.2 Exact constrained minimum over the admitted source

For $m \geq 0$, the original permitted changes of representative are

$$R_B u = R_{\text{ref}} u + \sum_{j=0}^m f_j(Bu)_j, \quad B : E \longrightarrow \mathbb{C}^{m+1}. \quad (\text{TB.3})$$

Set $N = d + m$. Denote by $\mathcal{P}_N$ the polynomials of degree at most N , with their coefficient columns in the monomial basis, and define the surjection

$$C_N : \mathcal{P}_N \longrightarrow E, \quad C_N P = j_h P. \quad (\text{TB.4})$$

Let H_N^ν be its actual moment Gram matrix:

$$(H_N^\nu)_{ij} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j d\nu_Z(t), \quad 0 \leq i, j \leq N. \quad (\text{TB.5})$$

It is positive definite by (TB.2). Define

$$K_N = C_N (H_N^\nu)^{-1} C_N^*. \quad (\text{TB.6})$$

This is positive definite: C_N is onto, so C_N^* is injective, and its nonzero vectors have positive quadratic form under $(H_N^\nu)^{-1}$.

Theorem 4.1 (The actual source minimum). *The orthogonally corrected representative R_m of PR 13 has the following explicit numerator and Gram matrix:*

$$p_{N,u} = (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad \mathcal{M} R_m u = v p_{N,u}, \quad (\text{TB.7})$$

$$\boxed{G_m = R_m^* R_m = U_\varepsilon^* K_N^{-1} U_\varepsilon.} \quad (\text{TB.8})$$

The polynomial correction

$$B_m(u)(s) = \frac{p_{N,u}(s) - R_Z(u)(s)}{h(s)} \quad (\text{TB.9})$$

has degree at most m . Its coefficients give exactly the map $B_m : E \rightarrow \mathbb{C}^{m+1}$ in the original source family (TB.3). Thus (TB.7) specifies the original theta boundary itself.

Proof. Taking Mellin transforms in (TB.3) gives

$$\mathcal{M} R_B u = v(s) \left(R_Z(u)(s) + h(s) \sum_{j=0}^m (Bu)_j s^j \right).$$

The expressions in parentheses range over precisely

$$\{P \in \mathcal{P}_N : C_N P = U_\varepsilon u\}. \quad (\text{TB.10})$$

Indeed the displayed expressions have the required residue. Conversely the difference between a polynomial in (TB.10) and $R_Z(u)$ is divisible by h , and the degree bound makes its quotient a polynomial of degree at most m . Mellin Plancherel identifies its squared Hilbert norm with $P^* H_N^\nu P$; this equality includes the measure $dt/(2\pi)$.

The vector $p = p_{N,u}$ in (TB.7) obeys $C_N p = U_\varepsilon u$. For $z \in \ker C_N$,

$$z^* H_N^\nu p = z^* C_N^* K_N^{-1} U_\varepsilon u = 0.$$

Therefore every competitor $p + z$ has the exact squared norm

$$(p + z)^* H_N^\nu (p + z) = u^* U_\varepsilon^* K_N^{-1} U_\varepsilon u + z^* H_N^\nu z. \quad (\text{TB.11})$$

Positivity proves uniqueness of the minimum. The original orthogonal projection construction is the unique minimum over the same affine source family, so its representative is (TB.7). Polarization proves (TB.8), while (TB.10) proves (TB.9). Every equality has been proved before application of the quotient; after applying q and J_Z one still has $qR_m = \sigma_Z$ and $J_Z R_m = I_E$. $\square$

The formula also extends to $m = -1$, $N = d - 1$, with no allowed source correction: C_{d-1} is the identity in the retained polynomial coordinates, and (TB.7) is precisely $R_Z(u)$.

4.3 Monic arithmetic polynomials and their exact recurrence

For this subsection use ν_Z , with no symmetry assumption on the selected packet. Let q_n be its monic polynomial of degree n orthogonal to all smaller degrees, and let $\kappa_n = \|q_n\|_{\nu_Z}^2 > 0$. The polynomials are explicitly the result of subtracting the projection of s^n onto smaller monomials. This is a specified unitriangular change of auxiliary polynomial basis; no coordinate of E , no arithmetic operator, and no polynomial coefficient has been replaced by an orthonormal coordinate.

Lemma 4.2 (The retained vertical-line recurrence). *There are $b_n \in \mathbb{C}$, with $b_n + \bar{b}_n = 1$, such that*

$$sq_0 = q_1 + b_0 q_0, \quad sq_n = q_{n+1} + b_n q_n - \frac{\kappa_n}{\kappa_{n-1}} q_{n-1} \quad (n \geq 1). \quad (\text{TB.12})$$

Proof. On the original line $s = 1/2 + it$, one has $\bar{s} = 1 - s$. Thus multiplication by s has adjoint multiplication by $1 - s$ on polynomial arguments. The coefficient of q_{n+1} in sq_n is one because both are monic. If $j \leq n - 2$, then

$$\langle q_j, sq_n \rangle_{\nu_Z} = \langle (1 - s)q_j, q_n \rangle_{\nu_Z} = 0,$$

since the first polynomial has degree at most $n - 1$. For $j = n - 1$ the same identity is $-\kappa_n$; division by κ_{n-1} gives the displayed negative coefficient. Finally

$$(b_n + \bar{b}_n)\kappa_n = \langle q_n, sq_n \rangle + \langle sq_n, q_n \rangle = \kappa_n.$$

This proves all coefficients, including their signs. No assertion that $b_n = 1/2$ is needed for a general packet. $\square$

Define the retained complete residue columns

$$v_n = j_h q_n \in E. \quad (\text{TB.13})$$

Changing the auxiliary monomial basis in (TB.6) to the stated monic basis and retaining its diagonal Gram matrix gives

$$K_N = \sum_{n=0}^N \frac{v_n v_n^*}{\kappa_n}. \quad (\text{TB.14})$$

Equivalently, this is the complete jet of the polynomial kernel

$$\mathcal{K}_N(z, w) = \sum_{n=0}^N \frac{q_n(z) \overline{q_n(w)}}{\kappa_n}. \quad (\text{TB.15})$$

In Taylor-jet coordinates at each centre, v_n has entries $q_n^{(k)}(\rho)/k!$ for $0 \leq k < m_\rho$. The map from these entries to the retained basis of E is the inverse confluent Vandermonde/Chinese-remainder map. In particular the factorials and all repeated-root derivatives remain in the construction.

Theorem 4.3 (Confluent kernel boundary formula). *For every $N \geq d - 1$,*

$$\boxed{AK_N + K_N A^* - K_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N}}. \quad (\text{TB.16})$$

Before taking any jets the corresponding polynomial identity is

$$(z + \bar{w} - 1) \mathcal{K}_N(z, w) = \frac{q_{N+1}(z) \overline{q_N(w)} + q_N(z) \overline{q_{N+1}(w)}}{\kappa_N}. \quad (\text{TB.17})$$

Proof. Apply j_h to (TB.12). The result replaces its left side by Av_n , because $j_h(sq_n) = A(j_h q_n)$. Substitute these relations into

$$AK_N + K_N A^* - K_N = \sum_{n=0}^N \frac{(Av_n) v_n^* + v_n (Av_n)^* - v_n v_n^*}{\kappa_n}.$$

The diagonal terms vanish by $b_n + \bar{b}_n = 1$. The negative contribution involving v_{n-1} has coefficient $-1/\kappa_{n-1}$, and cancels exactly the preceding positive contribution. The sole uncanceled term is the $n = N$ upper edge, giving (TB.16). The identical substitution before applying j_h proves (TB.17) for all complex z, w . This is a polynomial identity; no division by $z + \bar{w} - 1$ is performed at its zero set, and differentiating it proves the full confluent version. $\square$

4.4 Exact relative weight control from the same kernel

The original integration-by-parts form remains

$$W_m = A^* G_m + G_m A - G_m = -\{R_m^* (DR_m - R_m A) + (DR_m - R_m A)^* R_m\}. \quad (\text{TB.18})$$

The boundary term $[x \bar{F} H]_0^\infty$ vanishes for $F, H \in \mathcal{B}$. Put

$$L_N = AK_N + K_N A^* - K_N.$$

Commutation of A with U_ε in (TB.8) proves the exact comparison

$$\boxed{W_m = U_\varepsilon^* K_N^{-1} L_N K_N^{-1} U_\varepsilon}. \quad (\text{TB.19})$$

For clarity, multiplication gives

$$K_N^{-1} L_N K_N^{-1} = K_N^{-1} A + A^* K_N^{-1} - K_N^{-1},$$

which is exactly the middle factor in (TB.18). This proves the morphism between the inverse-kernel and original Hilbert metrics.

Define the actual real and complex numbers

$$\alpha_N = v_{N+1}^* K_N^{-1} v_{N+1}, \quad \beta_N = v_N^* K_N^{-1} v_N, \quad \gamma_N = v_N^* K_N^{-1} v_{N+1}. \quad (\text{TB.20})$$

Then the least symmetric allowance for the actual pair (G_m, W_m) is

$$\epsilon_m = \frac{|\operatorname{Re} \gamma_N| + \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}. \quad (\text{TB.21})$$

Here $-\epsilon_m G_m \preceq W_m \preceq \epsilon_m G_m$. To prove minimality, the invertible change $y = K_N^{-1} U_\varepsilon u$ turns its Rayleigh quotient into $y^* L_N y / (y^* K_N y)$. By (TB.16) the nonzero eigenvalues of this Hermitian comparison are the eigenvalues of the two-by-two matrix

$$\frac{1}{\kappa_N} \begin{pmatrix} \gamma_N & \beta_N \\ \alpha_N & \overline{\gamma_N} \end{pmatrix}.$$

Its trace and determinant give

$$\lambda_\pm = \frac{\operatorname{Re} \gamma_N \pm \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.$$

Cauchy-Schwarz in K_N^{-1} gives $|\gamma_N|^2 \leq \alpha_N \beta_N$, so the radicand is nonnegative; its square root is at least $|\operatorname{Re} \gamma_N|$. The maximum absolute eigenvalue is therefore (TB.21), including rank-one and zero cases.

For a packet closed under $\rho \mapsto 1 - \bar{\rho}$, with the multiplicities retained, $2 \operatorname{Re} \operatorname{tr} A = d$. Taking the trace in (TB.16) after multiplying by K_N^{-1} gives

$$0 = \operatorname{tr}(K_N^{-1} L_N) = \frac{2 \operatorname{Re} \gamma_N}{\kappa_N}. \quad (\text{TB.22})$$

Thus its real part is zero without any coordinate choice. Every eigenvector $Au = \rho u$ still satisfies

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{TB.23})$$

In particular (TB.21) has the intrinsic lower bound $\epsilon_m \geq \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|$. All of its quantities are now specified by arithmetic moments, complete jets, and the original local unit.

4.5 Exact change when one more theta direction is admitted

The step $m \mapsto m + 1$ means $N \mapsto N + 1$. Set $w = v_{N+1}$, $k = \kappa_{N+1}$, and $\theta = w^* K_N^{-1} w \geq 0$. Formula (TB.14) gives

$$K_{N+1} = K_N + \frac{w w^*}{k}. \quad (\text{TB.24})$$

Multiplication directly verifies the inverse identity

$$K_{N+1}^{-1} = K_N^{-1} - \frac{K_N^{-1} w w^* K_N^{-1}}{k + \theta}.$$

Consequently

$$G_{m+1} = G_m - \frac{U_\varepsilon^* K_N^{-1} w w^* K_N^{-1} U_\varepsilon}{k + \theta}, \quad (\text{TB.25})$$

$$\frac{\det G_{m+1}}{\det G_m} = \frac{k}{k + \theta} > 0. \quad (\text{TB.26})$$

The inverse formula follows by multiplying by (TB.24); If $w = 0$, both updates are identities and the determinant ratio is one. For $w \neq 0$, the determinant formula follows by extending $K_N^{-1} w$ to a basis, in which the rank-one map $K_N^{-1} w w^* / k$ has one possible nonzero eigenvalue θ/k . Hence the determinant factor of K_{N+1} relative to K_N is $1 + \theta/k$.

These are the same Hilbert metrics at adjacent relation levels. Subtract the two polynomial numerators in (TB.7) and divide their difference by h , as in (TB.9). This gives a source polynomial P_u of degree at most $m + 1$. The actual representative difference is $R_{m+1}u - R_m u = \Theta(P_u(D)\phi_*)$, an element of ΘW_{m+1} , where $W_{m+1} = \text{span}(\phi_*, D\phi_*, \dots, D^{m+1}\phi_*)$. The same source transition on the precise packet domain is $Q_{W_m, \Omega} \rightarrow Q_{W_{m+1}, \Omega}$, where $Q_{W, \Omega} = \mathcal{B}_\Omega / \Theta W$. Its kernel is $\Theta W_{m+1} / \Theta W_m$, identified through $[v] \mapsto [\Theta v]_{W_m}$ with the original one-dimensional space W_{m+1} / W_m . The unchanged target inclusion $\mathcal{B}_\Omega \hookrightarrow \mathcal{B}$ embeds this sequence in the original full-target quotient sequence. At an independent coefficient mask $\mathfrak{a}$, the kernel is exactly $(W_{m+1} / W_m)^\mathfrak{a}$, with its outer source and leg labels retained by (D12). The scalar e gives the zero at that same label; τ gives external absence. These spaces satisfy $DW_m \subseteq W_{m+1}$, so their generator is the actual adjacent-stage map (D14), not an assumed endomorphism of W_m . The full surviving primitive and the resulting arithmetic identity are proved in (TB.37)–(TB.39). None of these typed source maps changes the Grams in (TB.24)–(TB.26).

4.6 The original measure supplies a full tail recurrence

There is a second exact description which uses the unchanged measure μ of (TB.2). Let Q_n denote its monic orthogonal polynomials and $h_n = \|Q_n\|_\mu^2$. The reality and functional equation of g make this measure invariant under $s \mapsto 1 - s$; uniqueness of the monic orthogonal polynomial then proves $Q_n(1 - s) = (-1)^n Q_n(s)$. The recurrence proof above therefore gives

$$sQ_n = Q_{n+1} + \frac{1}{2}Q_n - \frac{h_n}{h_{n-1}}Q_{n-1} \quad (n \geq 1). \quad (\text{TB.27})$$

For $n = 0$, omit the last term.

Define the rational function-valued map and coefficient rows

$$r(u)(s) = \frac{R_Z(u)(s)}{h(s)}, \quad d_n u = \frac{1}{h_n} \int_{\mathbb{R}} \overline{Q_n(1/2 + it)} r(u)(1/2 + it) d\mu(t). \quad (\text{TB.28})$$

At an actual line zero a rational pole in this expression is interpreted in $L^2(\mu)$; its product with g is the entire function $vR_Z(u)$. In particular its norm and every displayed pairing are finite. Polynomials are dense in $L^2(\mu)$: the positive exponential moment from PR 13 makes the Fourier transform of an orthogonal annihilator analytic in a strip, and all its derivatives vanish at zero, so Fourier uniqueness annihilates it. Therefore Parseval gives

$$\mathcal{U}R_m u = g \sum_{n>m} Q_n d_n u, \quad \boxed{G_m = \sum_{n>m} h_n d_n^* d_n}. \quad (\text{TB.29})$$

The series is a norm-convergent positive matrix series.

Let $\ell_Z(u) = [s^{d-1}]R_Z(u)$ be the original rank-one extension coefficient. Polynomial division gives the exact identity $sr - rA = \ell_Z$. Pair it with Q_n , and use the adjoint relation $s^* = 1 - s$ and (TB.27). With $d_{-1} = 0$, the result is

$$\boxed{\frac{h_{n+1}}{h_n} d_{n+1} = d_{n-1} + d_n(\frac{1}{2}I - A) - \delta_{n0}\ell_Z}. \quad (\text{TB.30})$$

This computation uses only finite pairings of the rational function and polynomials; it does not differentiate an unproved termwise infinite expansion.

The polynomial correction in (TB.3) has last row $b_m = -d_m$, since its polynomial is $-\sum_{j=0}^m Q_j d_j$ and Q_m is monic. Furthermore $R_m^* f_{m+1} = h_{m+1} d_{m+1}^*$. Substitution in the exact PR 13 boundary contraction gives

$$\boxed{W_m = h_{m+1}(d_{m+1}^* d_m + d_m^* d_{m+1})}. \quad (\text{TB.31})$$

Thus the moment-kernel description (TB.16)–(TB.21) and the full tail description (TB.29)–(TB.31) calculate the same operator and metric through explicitly proved maps.

Theorem 4.4 (The full source defect and its observation of the packet). *The entire original derivative boundary, including its orthogonal lower source term, is*

$$\boxed{DR_m - R_m A = -\frac{h_{m+1}}{h_m} Q_m(D) f_0 d_{m+1} - Q_{m+1}(D) f_0 d_m.} \quad (\text{TB.35})$$

An exact primitive in V is obtained by replacing f_0 with ϕ_* in this formula. The finite observation

$$\mathcal{O}_m : E \longrightarrow \mathbb{C}^{2d}, \quad u \longmapsto (d_m A^k u, d_{m+1} A^k u)_{k=0}^{d-1} \quad (\text{TB.36})$$

is injective. In particular, its two retained source rows, together with the unchanged arithmetic action, distinguish every vector and every nilpotent jet in the original packet.

Proof. Use the finite expression $r_m = r - \sum_{j=0}^m Q_j d_j$ and compute

$$sr_m - r_m A = \ell_Z - \sum_{j=0}^m (sQ_j d_j - Q_j d_j A).$$

Insert (TB.27) and group coefficients of Q_j . For $0 \leq j < m$ the coefficient is zero by (TB.30); the $j = 0$ equation includes precisely the term ℓ_Z . The coefficient at Q_m is $-h_{m+1}d_{m+1}/h_m$, and that at Q_{m+1} is $-d_m$. Multiplying by g and applying the inverse Mellin map proves (TB.35). Its displayed primitive belongs to V , since every $D^j \phi_*$ does, and its theta image is the entire right side. Thus the term $Q_m(D) f_0$ remains an actual supported boundary even though its pairing with R_m vanishes.

Suppose that $\mathcal{O}_m u = 0$. Since $h(A) = 0$, polynomial division by h proves $d_m A^k u = d_{m+1} A^k u = 0$ for every $k \geq 0$. Equation (TB.30) for $n = m + 1$, applied to each $A^k u$, gives $d_{m+2} A^k u = 0$ for every k . Induction with the same recurrence gives $d_n A^k u = 0$ for all $n > m$ and $k \geq 0$. In particular every term of the positive series in (TB.29) vanishes on u , so $u^* G_m u = 0$. The exact left inverse $J_Z R_m = I_E$ makes G_m positive definite, hence $u = 0$. This proves injectivity without assuming distinct roots or removing a Jordan direction. $\square$

4.7 Metric dependence and the precise mathematical gain

Positivity in (TB.8) is the positivity of the original $L^2(dx)$ norm on an injective representative map. For the same-packet residue and dual-metric comparisons (TB.32)–(TB.33), assume explicitly that $Z = Z^\dagger$ under $\rho \mapsto 1 - \bar{\rho}$, with every full order retained. The preceding Christoffel-kernel and source-boundary results allow arbitrary packets. Reflection on a general packet is the conjugate-linear map $E_Z \rightarrow E_{Z^\dagger}$ defined in Section ?? . The original arithmetic residue matrix S , satisfying $S^* = -S$ and $A^* S + S A = S$, is still related to its dual metric by

$$G_m^D = S^* G_m^{-1} S, \quad W_m^D = -S^* G_m^{-1} W_m G_m^{-1} S. \quad (\text{TB.32})$$

The original arithmetic Weil form, with $J_g = M_{j_h, g'}$, is

$$u^* S J_g v = \langle u, (G_m^{-1} S J_g) v \rangle_{G_m}. \quad (\text{TB.33})$$

Thus the map relating the positive source metric to the arithmetic form is explicitly $G_m^{-1} S J_g$. A positive matrix G_m does not supply its positivity.

For each fixed packet, the original polynomial theta span is dense in $\mathcal{H}$, as proved in PR 13. Hence $G_m \rightarrow 0$. Equations (TB.8) and (TB.26) then imply

$$\lambda_{\min}(K_{d+m}) \longrightarrow +\infty, \quad \prod_{m=m_0}^M \frac{\kappa_{d+m+1}}{\kappa_{d+m+1} + v_{d+m+1}^* K_{d+m}^{-1} v_{d+m+1}} \longrightarrow 0. \quad (\text{TB.34})$$

Indeed $K^{-1} = U_\varepsilon^{-*} G_m U_\varepsilon^{-1} \rightarrow 0$, and the product telescopes to the positive determinant ratio. Absolute decay of the Gram matrix thus has a specified arithmetic-kernel counterpart. The relative quantity remains the exact expression (TB.21), with the spectral lower bound (TB.23).

The new result is a source-constrained, multiplicity-preserving arithmetic kernel formula and its exact finite relative control. It makes the next calculation depend on the concrete ν_Z -moments and the two retained complete jet vectors, rather than on freely selected positive matrices. No estimate that makes (TB.21) tend to zero for all actual packets has been proved here. A claim of spectral purity would require proving that estimate on these arithmetic quantities; (TB.23) identifies exactly what such a result would establish.

For standard background on orthogonal polynomial recurrence and kernel identities, see NIST DLMF, Section 18.2, <https://dlmf.nist.gov/18.2>. All identities used above, including the vertical-line signs, full jet maps, and metric comparisons, have been proved here for the retained objects.

4.8 Propagation through the original finite source tower

The finite spaces W_m used after (TB.26) have the precise source-domain maps (D14), because $DW_m \subseteq W_{m+1}$. Let $T_m = \Phi_m B_m : E \rightarrow W_m$, so $R_m = s_h + \Theta T_m$. Retain every outer leg label and every independent coefficient mask $\mathfrak{a} \subseteq I_n$. Then

$$\begin{aligned} 0 \rightarrow (W_{m+1}/W_m)^\mathfrak{a} &\rightarrow Q_{W_m, \Omega}^\mathfrak{a} \rightarrow Q_{W_{m+1}, \Omega}^\mathfrak{a} \rightarrow 0, \\ \sigma_{m, W_m}^\mathfrak{a} &= \sigma_{h, W_m}^\mathfrak{a}, \quad \sigma_{m+1, W_{m+1}}^\mathfrak{a} = \sigma_{m, W_{m+1}}^\mathfrak{a}, \end{aligned} \tag{TB.37}$$

where $\sigma_{m, W} u = [R_m u]_W$ and $A^\mathfrak{a}$ denotes the coordinatewise direct sum of the original packet operator A . The exact sequence is (D12); its kernel map is $[v_i] \mapsto [\Theta v_i]_{W_m}$, not an erasure of a coefficient face. The first section equality holds because $T_m(E) \subseteq W_m$. The second holds because the actual difference $R_{m+1} - R_m$ belongs to ΘW_{m+1} , as proved by the numerator difference following (TB.26).

The entire primitive in (TB.35), with f_0 replaced by ϕ_* , belongs to W_{m+1} . Hence (TC29b) and (TB.37) give the adjacent-level arithmetic identity

$$\overline{D}_{W_m, W_{m+1}} \sigma_{m, W_m}^\mathfrak{a} = \sigma_{m+1, W_{m+1}}^\mathfrak{a} A^\mathfrak{a}. \tag{TB.38}$$

This is an equality between the declared quotient spaces, obtained from the actual nonzero source boundary in (TB.35). It does not discard the lower source term or identify W_m with a D -invariant source. Before passing to W_{m+1} , that boundary retains its class $[K_{R_m}]_{W_m}$ in $(V/W_m)^\mathfrak{a}$. In particular, modulo W_m its exact remaining term is

$$[K_{R_m} u]_{W_m} = -[D^{m+1} \phi_*]_{W_m} d_m u, \tag{TB.39}$$

since Q_m has degree m and Q_{m+1} is monic. Linear independence of the source iterates shows that this displayed generator of W_{m+1}/W_m is nonzero. Thus the retained source transition measures exactly the row d_m , while the full primitive in W_{m+1} retains also d_{m+1} as in (TB.35). The injective two-row observation (TB.36) therefore remains compatible with, and more informative than, this one transition.

All maps here also commute with the original inclusion from $\mathcal{B}_U$ to $\mathcal{B}_\Omega$ when the packet lies in U . For a packet containing critical-line centres the codomain is the specified $\mathcal{B}_\Omega$, whose quotient over $\mathcal{B}_U$ is the full $(T_L Q)^\mathfrak{a}$ in (D13). The Gram formulas (TB.7)–(TB.8), rank-one updates (TB.25)–(TB.26), and full source formula (TB.35) are unchanged actual functions at these domains. This propagates the later mixed-face and critical-line precision through the original arithmetic source tower.